AnalytIQ Packet

Compliance, Security, and Vendor Documentation

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AnalytIQ Unified Learning Platform

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What AnalytIQ Is:

AnalytIQ is an AI-powered learning platform that turns traditional course materials into interactive, hands-on learning experiences. It combines lecture content, practice exercises, and guided AI assistance in one environment so students can move from reading to doing without switching tools. Learning is treated as active engagement: students read, practice, get feedback, and track progress in a single workflow.

Core Purpose:

The platform addresses a common gap in education: instructional content (lectures, slides) often stays separate from practice. AnalytIQ narrows that gap by turning static materials into structured learning activities. Uploaded content can become summaries, quizzes, and coding exercises that reinforce understanding in real time.

Who the Platform Serves:

- Students use AnalytIQ to access course materials, complete coding exercises in an

in-browser editor, request AI-guided hints, and monitor their progress. The experience is built to feel like a modern coding workspace rather than a traditional academic portal.

- Professors use AnalytIQ to create and manage courses efficiently: create classes,

upload or import content (PDFs or web sources), review AI-generated exercises, and monitor student performance through class and student-level analytics.

- Institutions use AnalytIQ as an integrated tool alongside their learning management

system (LMS). Via LTI 1.3, the platform supports single sign-on, roster synchronization, and optional grade passback (Assignment and Grade Services), without requiring separate account provisioning.

- Tenant and platform administrators can manage organizations (tenants), users, optional

per-tenant SSO (e.g. Google, Microsoft, OIDC), and platform-wide settings through dedicated admin capabilities.

Key Capabilities:

Content ingestion and structure:

- Lecture content can be uploaded as PDFs or imported from URLs, PyPI documentation,

OpenStax, LibreTexts, or OER Commons. Supported formats include PDF, TXT, CSV, Excel, Jupyter notebooks, and DOCX. Content is converted into structured chapters; PDFs (and, where applicable, imported content) can trigger automatic generation of practice exercises via AI.

Practice and feedback:

• Students complete coding exercises in an integrated in-browser editor. Submissions are graded server-side using sandboxed code execution with AI-generated unit test cases (5–8 per exercise). The execution environment uses subprocess isolation, memory limits, and import controls. Scores, accuracy, and time are stored on the platform. Students can request contextual hints from an AI assistant that is designed to guide rather than give direct answers.

Progress and analytics:

• Progress is tracked per student and per class. Students see their own completion and performance; professors see class-wide and per-student progress, completion rates, and leaderboards to identify where support is needed.

Resources:

• In addition to chapter-linked materials, professors can attach other resources (e.g. files, links) to a class for a single resource library experience.

Role of Artificial Intelligence:

AI in AnalytIQ is used as an educational support tool, not as an automated decision-maker. It is

used to:

• Generate practice content (e.g. exercises from chapter text).

• Provide learning assistance (e.g. chat and hints in the code-assistant and practice workflow). Student data is processed only to deliver these features and is not used to train AI models. AI interactions are powered by the Google Gemini API, with rate limits and safeguards in place.

Integration with Institutional Systems:

AnalytIQ is built to work alongside institutional systems, not replace them. Through LTI 1.3 and

LTI Advantage:

• Users can launch into AnalytIQ from their LMS (single sign-on; user and context identified from the LMS).

• Rosters can be synchronized via the Names and Roles Provisioning Service (NRPS).

• Optional grade passback (Assignment and Grade Services) allows scores to be sent to the LMS gradebook; academic records and official grades remain under institutional control. The platform can be used standalone (email/password signup and sign-in, optional per-tenant SSO) or as an LTI tool inside an LMS.

Overall Platform Vision:

AnalytIQ aims to provide a unified learning environment where instructional content, interactive practice, and AI support live in one system. By reducing the gap between teaching and application, the platform is designed to improve engagement, support skill development, and help both educators and learners achieve better outcomes.

Executive Summary:

AnalytIQ is a web-based learning platform for data analytics and coding education. It combines structured course delivery, hands-on practice, and AI-powered support in one place. The platform is used through distinct but connected experiences: a Professor Portal for creating and managing coursework, a Student Portal for learning and practicing, optional LTI-based access from an LMS, and administrative tools for tenants and platform-wide settings. The platform closes the gap between theory and practice by turning lecture PDFs (and other imported content) into interactive coding exercises and quizzes, with a context-aware AI tutor available inside the workflow.

1. Purpose and Target Audience

Platform objective:

To provide an ecosystem for analytics and coding education that:

- Helps professors digitize curriculum quickly, generate practice material from uploaded or imported content, and monitor student performance.
- Helps students learn from course content, practice with real exercises, and get guided help when stuck.
- Helps institutions deploy the tool inside their existing LMS using LTI 1.3, with single sign-on, roster sync, and optional grade passback.

Target users:

- University professors and educators who want to use AI-assisted tools to build and manage courses.
- University students in analytics, business, or coding-related courses.
- Self-learners and professionals building applied analytics and coding skills (when given access, e.g. via a class code or LMS link).

2. Core Platform Capabilities:

2.1 Professor and Student Experiences:

Professor experience:

• Class management: Create classes (each with a unique class code), manage chapters, and organize materials.

• Content upload and import: Upload lecture PDFs (and other allowed file types) or import from web sources (URL, PyPI, OpenStax, LibreTexts, OER Commons) via a unified “Import from Web” flow.

• Content review: Review and edit AI-generated exercises and chapters before students see them; add or edit exercises manually.

• Student analytics: View class roster, per-student and class-wide progress, completion, and leaderboards; identify common difficulty points.

• Resources: Attach and manage class resources (e.g. PDFs, datasets, links) alongside chapters.

• LTI setup (optional): Configure LTI 1.3 for a class (platform registration, deep linking, NRPS, optional AGS grade passback) via the LTI setup UI.

Student experience:

• Class enrollment: Join a class using the class code shared by the professor, or enter via an LTI launch from the LMS.

• Curriculum access: View chapters and read lecture/content (PDF or HTML) in the platform.

• Practice workflow: Complete coding exercises generated from or attached to class material in an in-browser editor.

• Progress tracking: See completion status and performance over time for their own work.

2.2 Learning Infrastructure:

Chapter modules:

• Chapters are learning units tied to a class; each can have a title, description, and associated file (e.g. PDF) or imported content.

• Lecture/content is viewable in-browser (PDF or HTML).

• Progress is tracked at the exercise level and aggregated for the student and for the professor’s view.

Practice exercises:

• Auto-generated: Created from chapter text (e.g. PDF extraction) using the Gemini API; professors can enable or disable this on upload/import.

• Instructor-authored: Professors can add and edit exercises (title, description, difficulty, solution code) per chapter.

• Interactive coding: Exercises support Python, SQL, R, JavaScript, and Java, with automatic language detection from syllabus content. Students complete exercises in an embedded in-browser editor (syntax highlighting, line numbers).

• Tracking: Submissions are evaluated server-side via sandboxed execution against AI-generated test cases; score, accuracy, and time are stored in the practice_progress model. Completion and best scores drive progress views and leaderboards.

2.3 AI-Powered Learning Support

AnalytIQ provides an AI assistant that supports learning while keeping the student in control of their work.

Capabilities:

• Explain concepts and answer questions in a chat interface (e.g. code-assistant).

• Help debug errors and guide students toward solutions.

• Provide hints in context of the current exercise and code (where integrated).

• Support professors by generating exercises (and optionally quizzes/summaries) from uploaded or imported content. Implementation

• Powered by the Google Gemini API (e.g. configurable model such as gemini-1.5-flash).

• Rate limits (e.g. per user per minute/hour) and message/size caps apply to chat.

• No use of student data for model training.

2.4 Progress and Analytics

For professors:

• Class-level dashboards (e.g. student count, chapter and exercise counts).

• Per-class progress view: which students have completed which exercises, scores, and aggregates.

• Leaderboards (e.g. by class, by exercise, grand leaderboard) to spot bottlenecks and high performers.

For students:

• Personal progress across enrolled classes and exercises.

• Visibility into completed and pending activities and performance.

2.5 Resource Library

Professors can attach resources to a class (e.g. lecture PDFs, datasets, links). Resources are listed with the class and can be downloaded or opened with appropriate handling (e.g. PDF viewing, file download). Chapter-linked content and class resources together form the course material and resource library.

2.6 Quality and Continuous Improvement

AnalytIQ instruments exercise generation events, grading outcomes, and computed quality scores per exercise to monitor and improve content quality over time. Prompt template versioning enables controlled evaluation of AI generation approaches. This data is used only for platform improvement and is not shared externally.

3. How Learning Happens in AnalytIQ:

Typical flow

1. Professor creates a class and receives a unique class code.

2. Professor adds a chapter (upload PDF or import from web) and may enable automatic exercise generation from the content. 3. Platform extracts text (and, where applicable, imports content), optionally calls the AI to generate exercises, and stores chapters and exercises. 4. Professor reviews and edits chapters and exercises, then publishes (students see content when they open the class). 5. Students enroll via the class code (or enter via LTI from the LMS), open the class, then open a chapter.

6. Students read the material in-browser and complete exercises in the practice

playground. 7. When stuck, students can request AI hints; they submit code and receive immediate feedback (server-side grading via sandboxed execution); scores are saved to the platform.

8. Professors review progress and leaderboards to see where support is needed.

9. Where LTI and AGS are configured, the platform can send grades to the LMS

gradebook.

4. Technical Architecture (High Level)

Technology stack

• Frontend: HTML, CSS, JavaScript (ES6+). Pages are server-rendered from Jinja2 templates (templates/), with interactive behavior in vanilla JS. No React/Vue.

• Backend: Python, Flask. Logic is split across blueprints (auth, users, classes, chapters, exercises, AI, health, LTI, superadmin, content import, platform, SSO).

• Database: PostgreSQL (e.g. Supabase). Access via connection pool (e.g. psycopg2); parameterized SQL only.

• Storage: Supabase Storage for chapter files and assets; presigned URLs for uploads and downloads.

• AI: Google Gemini API for chat and for exercise generation from content.

• Auth: JWT (Bearer token); optional per-tenant SSO (OAuth2/OIDC) with state in Redis.

• Async (optional): Celery with Redis for heavy workflows (e.g. large PDF processing) when the feature is enabled.

Codebase architecture:

The platform codebase is organized into a clean package layout: core configuration and shared utilities, service-layer business logic, background workers, 13 Flask blueprints for API routing, and 8 repository modules for database access. Schema changes are managed through versioned SQL migrations (10 migrations to date).

Key engineering points:

• PDF (and multi-format) to text, then to exercise generation pipeline.

• Role-based access (student, professor, tenant_admin, superadmin) enforced in the backend.

• Secure handling: parameterized queries, bcrypt password hashing, JWT validation, optional CSRF when enabled.

• LTI 1.3 readiness: OIDC login/launch, NRPS roster sync, optional AGS grade passback.

• Caching (e.g. Redis) and rate limiting for API and AI endpoints.

5. Security and Reliability (Summary)

• Authentication via JWT (and optional SSO); session state in token and localStorage; 401 handling redirects to login and clears tokens.

• Role-based access control so students see only their data and enrolled classes; professors see their classes and students; tenant admins and superadmins have scoped administrative access.

• Encryption in transit (HTTPS) and at rest via the hosting and database provider.

• Operational logging (e.g. request IDs, structured logs), health checks, and optional error tracking (e.g. Sentry) for reliability and debugging.

• Middleware for IP blocklist and maintenance mode; superadmin can bypass maintenance where configured.

6. User Experience Highlights

• Clear separation between professor and student workflows (different pages and API access by role).

• Smooth flow from reading (chapter view) to practicing (playground) without leaving the platform.

• Embedded coding playground with immediate feedback and optional AI hints.

• AI help integrated into the learning loop (chat and hints) without replacing student work.

7. Future Enhancements (Conceptual)

• Your document's future ideas (study groups, more advanced automated grading, video support, deeper LMS reporting) can remain as-is; the following aligns with how the

platform works today:

• Grading: Exercise grading uses server-side sandboxed code execution with AI-generated unit tests. Scores are stored and can be passed to the LMS via AGS. Future work may include expanded grading logic for additional exercise types.

• LMS: LTI 1.3 and NRPS are implemented; AGS is optional per class. Deeper institutional analytics would build on this.

Simple Summary AnalytIQ helps professors turn PDFs and other content into interactive coursework (chapters + exercises) and helps students learn by reading, practicing, and getting AI guidance in one place. It works standalone (email/password and optional tenant SSO) or inside an LMS through LTI 1.3 with single sign-on, roster sync, and optional grade passback.

Terms and Conditions

Terms and Conditions

Last Updated: February 20, 2026

These Terms and Conditions (“Terms”) govern your access to and use of the AnalytIQ platform and related services (the “Service”) operated by AnalytIQ (“AnalytIQ,” “we,” “our,” or “us”). By creating an account, accessing, or using the Service, you agree to be bound by these Terms.

1. Description of the Service

AnalytIQ provides a web-based platform to support teaching and learning in data analytics and related fields.

The Service may include:

• Course and class management — Creating and managing classes, chapters, and resources; student enrollment via class code or through integration with an institutional learning management system (LMS).

• Content creation and import — Uploading course materials (e.g. PDFs and other supported formats) or importing content from permitted web sources (e.g. URLs, documentation, or educational libraries) to create structured learning modules.

• Practice exercises and progress — Interactive coding exercises (including those generated from course content), in-browser practice, automated scoring and progress tracking, and analytics for instructors.

• AI-powered learning assistance — Chat and contextual hints to support learning; AI is used only to deliver these features and not to train models on your data.

• Integration with institutional learning management systems — Launch from the LMS, roster synchronization, and optional grade reporting (LTI 1.3), where the institution has enabled such features.

• User account and profile management — Registration, sign-in (including optional single sign-on where offered), profile and preference settings, and account deletion. AnalytIQ may modify, suspend, or discontinue any feature of the Service at any time.

2. Account Registration and Security

You must provide accurate and complete information when creating an account.

You are responsible for:

• Keeping your login credentials confidential

• All activity that occurs under your account

You agree to notify AnalytIQ promptly of any unauthorized access or suspected security breach.

3. Acceptable Use

You agree not to:

- Violate any applicable laws or regulations
- Harass, threaten, or harm other users
- Attempt unauthorized access to systems, accounts, or data
- Use automated tools to scrape or extract data without permission
- Upload unlawful, infringing, or harmful content
- Interfere with platform functionality or security (e.g. circumventing rate limits or access

controls) AnalytIQ may suspend or terminate accounts that violate these Terms or for security or operational reasons.

4. User Content

You retain ownership of content you upload or submit to the platform (e.g. course materials, exercises, or other resources). By submitting content, you grant AnalytIQ a limited, non-exclusive license to store, process, and use that content only to operate and provide the Service (e.g. displaying it to enrolled users, generating exercises where applicable, storing it in our systems). You represent that you have the rights necessary to upload and share such content and that doing so does not violate any third-party rights or applicable law.

5. Intellectual Property

All platform software, designs, documentation, and materials provided by AnalytIQ are protected by intellectual property laws and remain the property of AnalytIQ or its licensors. You may not reproduce, distribute, or commercially exploit platform materials except as needed for normal use of the Service under these Terms.

6. AI Powered Features

AnalytIQ includes AI-powered tools (e.g. chat and hints) intended to assist learning and educational workflows.

You acknowledge that:

- AI-generated outputs are for educational assistance only and may be incomplete or

inaccurate.

- They should not be relied upon as professional, legal, or authoritative advice.
- You remain responsible for evaluating and verifying any AI-generated content.

AI prompts and responses are used only to provide the feature (e.g. conversation continuity). They are not used to train AI models. Use of AI features may be subject to rate limits and usage policies

7. Instructor and Student Relationships

AnalytIQ acts solely as a technology provider. AnalytIQ does not control:

- Course quality or content decisions
- How grades or progress are used for academic purposes
- Academic or institutional policies
- Institutional standards or accreditation

Those responsibilities remain with the relevant educational institution, instructors, and students.

8. Privacy

Use of the Service is also governed by the AnalytIQ Privacy Policy, which describes how personal data is collected, used, and protected. By using the Service, you acknowledge that you have read and accept the Privacy Policy. Compliance and Data Protection Provisions

9. Education Records and FERPA Compliance

Where AnalytIQ provides services to educational institutions, it does so as a “school official” with a “legitimate educational interest” under the U.S. Family Educational Rights and Privacy Act (FERPA).

AnalytIQ:

- Processes education records solely on behalf of institutions and in accordance with documented instructions.
- Does not own student education records.
- Uses student data only for educational and operational purposes of the Service.
- Does not use education records for advertising or unrelated commercial purposes.

Ownership and control of education records remain with institutions and students as applicable.

10. Institutional Control of Data

Where the Service is used under an institutional agreement, educational institutions act as data controllers and AnalytIQ acts as a data processor.

Student data is processed:

- Only under institutional direction and in accordance with the Data Processing Agreement and applicable law.
- Solely to deliver the educational services and related operations.

Requests to access, correct, or delete education records in the institution’s capacity as controller should be directed to the relevant institution. AnalytIQ will assist the institution as required by contract or law.

11. AI Data Use Restrictions

AnalytIQ does not use student data, coursework, or submissions to train artificial intelligence models.

Data processed by AI systems (e.g. prompts and responses):

- Is used only to provide the requested functionality (e.g. chat, hints).
- Is not used for model training.
- May be stored as needed for service continuity (e.g. multi-turn conversation); such storage is not for training.
- Is not shared for advertising or external commercial use.

12. International Data Transfers

AnalytIQ is based in the United States and uses infrastructure located in the United States. If you are located outside the United States, you consent to the transfer and processing of your data in the United States in connection with the Service.

13. LMS Integration and Academic Records

AnalytIQ may integrate with institutional learning management systems (e.g. via LTI 1.3) for launch, roster sync, and optional grade reporting. AnalytIQ is not the official system of record for academic grades. Final academic records and grading decisions remain under institutional control. Where grade passback is used, the institution's LMS is the system of record for those grades. Legal Protections

14. Disclaimer of Warranties

The Service is provided on an "as is" and "as available" basis. To the fullest extent permitted by law, AnalytIQ disclaims all warranties, whether express or implied, including any implied warranties of merchantability, fitness for a particular purpose, or non-infringement.

15. Limitation of Liability

To the fullest extent permitted by law, AnalytIQ shall not be liable for any indirect, incidental, consequential, special, or punitive damages (including loss of data, revenue, or profits) arising from or related to the Service or these Terms. Total liability of AnalytIQ for any claims arising from or related to the Service or these Terms

shall not exceed the greater of:

- The amount you (or the institution on whose behalf you use the Service) paid to AnalytIQ for the Service in the 12 months preceding the claim, or
- One hundred U.S. dollars (USD $100)

Some jurisdictions do not allow the exclusion or limitation of certain damages; in those jurisdictions, the above limitations apply only to the maximum extent permitted by law.

16. Indemnification

You agree to indemnify, defend, and hold harmless AnalytIQ and its officers, directors, employees, and agents from and against any claims, damages, losses, liabilities, and expenses (including reasonable attorneys' fees) arising from or related to:

- Your violation of these Terms
- Your misuse of the Service
- Your infringement of any third-party right (including intellectual property or privacy)
- Content you upload or submit through the Service

17. Termination

- You may stop using the Service and may request account deletion at any time, subject to

any institutional or contractual arrangements.

- AnalytIQ may suspend or terminate your access at any time for violations of these Terms

or for security or operational reasons, with or without notice where permitted by law. Upon termination, your right to access and use the Service ceases. Provisions that by their nature should survive (e.g. Sections 5, 8–13, 14–16, 18–20) will survive termination.

18. Governing Law

These Terms are governed by the laws of the Commonwealth of Massachusetts, United States, without regard to its conflict-of-laws principles. Any dispute arising from or related to these Terms or the Service shall be subject to the exclusive jurisdiction of the state and federal courts located in Massachusetts, to the extent permitted by applicable law.

19. Changes to Terms

AnalytIQ may update these Terms from time to time. We will indicate the "Last Updated" date at the top of the Terms. Material changes may be communicated via the Service or by other reasonable means. Continued use of the Service after the effective date of revised Terms constitutes your acceptance of the revised Terms. If you do not agree, you must stop using the Service. For institutional agreements, changes may be subject to the terms of the agreement with the institution.

20. Contact Information

For questions about these Terms, please contact AnalytIQ through the contact information provided on the official website (e.g. www.myanalytiq.org) or at techsupport@myanalytiq.org or aelharouchi@myanalytiq.org.

AnalytIQ Privacy Policy

AnalytIQ Privacy Policy

Last Updated: February 20, 2026

AnalytIQ operates the AnalytIQ learning platform and related services (the “Service”). This Privacy Policy explains how we collect, use, disclose, and protect information when you use the Service. This policy is designed to align with applicable privacy and education data protection laws, including the U.S. Family Educational Rights and Privacy Act (FERPA), applicable U.S. state privacy laws such as the California Consumer Privacy Act (CCPA/CPRA), and relevant international data protection principles. By using the Service, you agree to the practices described in this policy.

1. Information We Collect

1.1 Information You Provide

Account information We collect information you provide when registering or signing in, such as your name, email address, and account credentials. Passwords are stored in securely hashed form (e.g. bcrypt); we do not store plain-text passwords. We also store your account role (e.g. student, instructor, tenant administrator, or platform administrator), which may be set automatically (e.g. from your email domain or institution) or by an administrator. If you sign in through an institution’s single sign-on (SSO) or learning management system (LMS), we may receive and store identifiers from that provider (e.g. SSO or LTI subject identifier) to link your account. Profile information We may store optional profile details such as display name, biography, and profile image if you provide them through the Service. Class and course data We store information related to courses and classes (e.g. class name, description, class code), enrollments, chapters and exercises, and educational content you upload or create (e.g. PDFs, imported content, resources). This includes metadata and, where applicable, files stored in our systems. Learning and practice data

We collect data related to your use of the Service, including progress tracking, exercise results, scores, completion status, and time spent. Exercise code is submitted to the server for grading via sandboxed execution. We store the resulting scores and related metrics. Code submitted for grading is processed server-side and not retained beyond the grading operation, except where an instructor submits solution code as part of exercise content. AI feature data When you use AI features (e.g. chat or hints), we process your prompts, responses, and related context to provide the feature. This data may be stored in our systems to support multi-turn conversation and continuity. It is used only to deliver the AI functionality and is not used to train artificial intelligence models. Support and contact We collect information you provide when contacting us (e.g. through the platform’s contact form or support channels), such as your name, email, and message content.

1.2 Information Collected Automatically

Technical and usage information We collect technical information that is necessary for operations, security, and performance, such as IP address, request identifiers, and error or security-relevant events. This information is captured in application logs and, where deployed, in error-monitoring systems (with personal data minimized or filtered before being sent to third-party monitors). We do not currently use product analytics tools (e.g. feature clicks, session replay, or behavioral tracking) for advertising or cross-site profiling. Local storage and session data The platform uses browser local storage (e.g. localStorage) to maintain session state (e.g. authentication token, current user details) and user preferences (e.g. theme) so you can stay signed in and use the Service. We do not use this storage for advertising tracking.

1.3 Information from Service Providers

We use third-party service providers that process data on our behalf to operate the Service,

including:

• Supabase — Database (PostgreSQL) and object storage for course files and other stored content.

• Render (or equivalent) — Application hosting and, where used, Redis for rate limiting, caching, and (when SSO is enabled) OAuth state.

• Google Gemini API — AI-powered features (e.g. chat, hints, exercise generation).

• Email provider (e.g. Resend) — Transactional communications (e.g. password reset, security-related notices).

• Error and performance monitoring (e.g. Sentry) — With contractual and technical measures to limit exposure of personal data (e.g. PII filtering). Authentication for the Service (sign-up, sign-in, password reset, session tokens) is handled by AnalytIQ; we do not use Supabase or another third party as the primary authentication provider for user accounts. We do not currently use product analytics providers (e.g. PostHog) for usage analytics. Any future use of such tools would be disclosed in this policy and would not be used for advertising or cross-site behavioral profiling. These providers process data only as necessary to operate the Service and are contractually required to protect information in line with our instructions and applicable law.

2. How We Use Information

We use collected information to:

• Provide, operate, and maintain the Service.

• Authenticate users and manage accounts (including optional SSO and LMS-based access).

• Deliver educational functionality (e.g. classes, chapters, exercises, progress tracking, leaderboards).

• Provide AI-powered learning assistance (e.g. chat, hints) and, where applicable, content generation (e.g. exercises from course material).

• Send service-related communications (e.g. password reset, important security or account notices).

• Monitor system performance and security (e.g. logs, audit trail, error monitoring).

• Comply with legal obligations and enforce our Terms and Conditions.

AnalytIQ does not use personal information for advertising or for selling data to third parties.

3. Education Records and FERPA Compliance

When the Service is used by or on behalf of educational institutions, we act as a “school official” with a “legitimate educational interest” under the U.S. Family Educational Rights and Privacy Act (FERPA). Education records may include enrollment data, coursework, progress tracking, scores, and instructor-visible analytics. AnalytIQ processes education records solely on behalf of institutions and under their direction, in accordance with our Data Processing Agreement and applicable law.

AnalytIQ does not own student education records. Ownership and control remain with the educational institution and the student as applicable. Student data is never sold and is not used for advertising or unrelated commercial purposes.

4. Data Sharing

AnalytIQ does not sell personal information.

We may share data only in limited circumstances:

• With service providers that are necessary to operate the platform (as described in Section 1.3), under contractual safeguards.

• With instructors and authorized institutional users for educational purposes (e.g. viewing progress and analytics for their classes).

• When required by law or legal process (e.g. subpoena, court order, or applicable regulation).

• In connection with business transfers (e.g. merger, acquisition, or sale of assets), subject to the same privacy commitments where legally required. All service providers are required to protect data through contractual and, where applicable, technical safeguards.

5. Analytics and Tracking

We use application and security logs and, where deployed, error and performance monitoring to support operations, security, and troubleshooting. These do not constitute product analytics or behavioral tracking for advertising. We do not currently use product analytics tools (e.g. feature-level usage, session replay, or cross-site tracking) to collect usage patterns for advertising or profiling. Any future use of such tools would be described in this policy and would not be used for advertising or cross-site behavioral profiling. AnalytIQ does not use tracking technologies for advertising or cross-site behavioral profiling.

6. Data Retention and Deletion

We retain personal information only as long as necessary to provide the Service and to meet legal or contractual obligations (e.g. audit, legal hold), as described in our Data Retention and Deletion Policy. If you or your institution submits an account deletion request (or deletion is required under contract), we will remove personal data from active systems within a reasonable timeframe,

consistent with our technical capabilities and legal obligations. Content you created (e.g. classes, chapters) may be handled in accordance with our product behavior and any institutional or contractual terms. Educational institutions may retain their own copies of academic records in their systems (e.g. LMS); this policy does not govern institutional retention. System logs (including audit logs) are retained in accordance with our retention policy and the retention practices of our hosting and infrastructure providers.

7. Data Security

AnalytIQ implements safeguards designed to protect information, including:

- Encrypted data transmission — All access to the Service uses HTTPS (TLS).
- Encryption of stored data — Data at rest is protected using encryption provided by our

database and storage infrastructure; sensitive configuration (e.g. SSO secrets) is additionally encrypted by the application.

- Role-based access controls — Users can access only the data appropriate for their

role (e.g. student, instructor, administrator).

- Authentication and session protections — Passwords are hashed (e.g. bcrypt);

sessions use signed, expiring tokens (JWT); optional SSO and LTI use validated state and signatures.

- Secure cloud infrastructure — We use providers that maintain physical, network, and

operational security for their services. No system can guarantee absolute security. We work to protect your information and will notify affected institutions and users as required by law in the event of a breach involving personal data.

8. Artificial Intelligence Data Handling

AI features process your inputs (e.g. prompts, context) only to generate real-time responses and to support conversation continuity where the feature requires it. AnalytIQ does not use student data, coursework, or prompts to train artificial intelligence models. Any storage of AI-related data (e.g. conversation history) is for platform functionality only and is subject to the same retention and deletion practices as other user data.

9. International Data Transfers

AnalytIQ operates from the United States and hosts data on infrastructure located in the United States.

If you are located outside the United States, you consent to the transfer and processing of your information in the United States in connection with the Service. We take steps consistent with applicable law to protect your information in connection with such transfers.

10. User Rights

Depending on applicable law (e.g. CCPA/CPRA, GDPR where it applies), you may have rights

to:

- Access personal data we hold about you.
- Correct inaccurate information.
- Request deletion of personal data.
- Object to or restrict certain processing, where the law provides.
- Data portability, where required by law.

To exercise these rights, contact us using the information in Section 14. Students using the platform through an educational institution may need to direct requests regarding education records to their institution first; we will assist the institution as required by contract or law.

11. Children's Privacy

The Service is not directed at individuals under 18 years of age, and AnalytIQ does not knowingly collect personal information from minors. If we become aware that we have collected personal information from a minor without appropriate consent, we will take steps to delete that information in accordance with applicable law.

12. California Privacy Rights

If you are a California resident, you may have rights under the California Consumer Privacy Act

(CCPA/CPRA) and related laws, including the right to:

- Request access to the categories and specific pieces of personal information we have

collected about you.

- Request deletion of your personal information, subject to certain exceptions.
- Understand the categories of sources and purposes of collection.
- Non-discrimination for exercising your privacy rights.

AnalytIQ does not sell personal information and does not share personal information for cross-context behavioral advertising. To submit a request or ask questions about our practices, contact us using the information in Section 14.

13. Changes to This Policy

AnalytIQ may update this Privacy Policy from time to time. We will indicate the "Last Updated" date at the top. Material changes may be communicated through the Service or by other reasonable means. Continued use of the Service after the effective date of an updated policy constitutes acceptance of the updated policy. If you do not agree, you should stop using the Service.

14. Contact Information

For privacy-related inquiries, or to exercise your rights under this policy or applicable law, please contact AnalytIQ through the contact information provided on our official website (e.g. www.myanalytiq.org) or at techsupport@myanalytiq.org or aelharouchi@myanalytiq.org.

Data Processing Agreement (DPA)

Data Processing Agreement (DPA)

This Data Processing Agreement ("DPA") forms part of the agreement between AnalytIQ ("Processor") and the educational institution ("Controller"), together the "Parties".

1. Scope and Purpose

This DPA governs the processing of personal data (including student and staff records) that the Controller makes available to the Processor in connection with the use of the AnalytIQ learning platform. Personal data processed may include: identifiers (e.g. name, email address), authentication data (e.g. hashed passwords or SSO identifiers), role and institutional context, class enrollments, progress and assessment data (e.g. exercise submissions, scores, completion status), and other data generated or submitted in the course of using the platform (e.g. AI chat messages and preferences). The Processor processes such data only as necessary to provide the contracted learning and teaching services and related technical operations.

2. Processor Obligations

AnalytIQ agrees to:

• Process personal data only on documented instructions from the Controller (and, where the service is used under an institutional agreement, in accordance with that agreement and the platform's documented functionality).

• Implement appropriate technical and organizational measures to protect personal data, in line with Section 6.

• Ensure that personnel and any subprocessors authorized to process personal data are bound by confidentiality and data protection obligations.

• Assist the Controller in responding to data subject rights (e.g. access, correction, deletion, portability) to the extent feasible within the platform's capabilities and applicable law.

• Notify the Controller without undue delay after becoming aware of a confirmed personal data breach affecting the Controller's data.

3. Educational Records (FERPA)

For Controllers subject to the U.S. Family Educational Rights and Privacy Act (FERPA), AnalytIQ acts as a school official with a legitimate educational interest in the student records provided.

AnalytIQ processes such data solely to provide the contracted educational services and will not

use student data for:

• Marketing or advertising

• Commercial resale

• Training of artificial intelligence models

AI features (e.g. exercise generation and the in-platform tutoring/chat) are used to deliver the service; student inputs are not used to train third-party or AnalytIQ AI models.

4. Subprocessors

The Controller authorizes AnalytIQ to use existing and future subprocessors necessary to

operate the platform, including:

- Application and database hosting
- Object and file storage
- AI service providers used to deliver platform functionality (e.g. exercise generation and tutoring)
- Email delivery
- Caching and rate-limiting infrastructure
- Error and performance monitoring (with appropriate restrictions on use of personal data)

Such subprocessors will be subject to contractual obligations consistent with applicable data protection law. AnalytIQ remains responsible for ensuring subprocessors comply with those obligations to the same standard as under this DPA. A list of current subprocessors (and categories of processing) will be made available to the Controller on request and updated when material changes occur.

5. Return or Deletion of Data

Upon termination of the services or upon written request by the Controller, AnalytIQ shall, at the Controller's choice and to the extent technically and legally feasible:

- Delete all personal data processed on behalf of the Controller, or
- Securely return such data to the Controller in a commonly used, machine-readable format, unless applicable law requires or permits continued storage (e.g. legal hold, statutory retention). Any retained data will continue to be protected in accordance with this DPA.

6. Data Security Measures

AnalytIQ maintains technical and organizational measures appropriate to the nature of the data

and the risks, including:

- Encrypted data transmission: All access to the platform is over TLS (HTTPS); no transmission of personal data in clear text over public networks.
- Encryption at rest: Personal data at rest is stored in secure cloud infrastructure that provides encryption at rest (e.g. database and object storage).
- Access controls and authentication: Access to systems that process personal data is restricted. The platform uses strong authentication (e.g. password hashing with bcrypt, JWT-based session tokens, and optional SSO). Access to data is enforced by role-based rules (e.g. students, professors, tenant and platform administrators) so that users only access data they are authorized to see.
- Application security: Input is validated and parameterized queries are used to protect against injection; security headers are applied; and rate limiting and operational safeguards help protect availability and integrity.
- Audit and operations: Certain processing activities are logged for security and

operational purposes; logs are retained and accessed under controlled conditions.

- Operational safeguards: Procedures for access management, incident response, and

(where applicable) maintenance or availability modes, with due regard to data protection. These measures may be updated over time provided they remain at least equivalent in effectiveness to those described above.

7. Duration

This DPA remains in effect for as long as AnalytIQ processes personal data on behalf of the Controller under the main agreement between the Parties. Obligations that by their nature should survive (e.g. confidentiality, return/deletion, and audit rights) will continue after termination to the extent required by law or the main agreement. Signatures For the Data Controller (Institution) For the Data Processor (AnalytIQ)

Signature: _____ Signature: _____

Name and Title: _____ Name and Title: _____

Date: _____ Date: _____

AnalytIQ FERPA Compliance Statement

AnalytIQ FERPA Compliance Statement

Last Updated: February 20, 2026

1. Overview

AnalytIQ is committed to protecting the privacy and security of student education records in accordance with the Family Educational Rights and Privacy Act (FERPA), 20 U.S.C. § 1232g, and its implementing regulations. This statement describes how AnalytIQ supports educational institutions in maintaining FERPA compliance when using the AnalytIQ platform.

2. Role Under FERPA

When providing services to educational institutions, AnalytIQ operates as a school official with a legitimate educational interest under FERPA.

AnalytIQ qualifies as a “school official” because:

• It performs institutional services on behalf of schools (e.g. course delivery, practice exercises, progress tracking, optional LMS integration).

• It is under the direct control of the institution regarding the use of education records (AnalytIQ acts as a data processor and processes data only on documented instructions from the institution).

• It uses student data only for educational and operational purposes of the platform.

• It is subject to strict confidentiality and data protection obligations under its agreements and policies. AnalytIQ processes student information solely as directed by the institution and in accordance with its Data Processing Agreement and this statement.

3. Education Records Processed by AnalytIQ

AnalytIQ may process education records that include:

• Student account information (e.g. name, email, role, authentication-related data).

• Enrollment and roster data (e.g. class enrollments, and when LTI is used, roster data from the institution’s LMS).

• Coursework and content (e.g. classes, chapters, exercises, and related materials created or assigned).

• Exercise performance and progress (e.g. scores, completion status, time spent, and similar metrics; exercise code is graded server-side via sandboxed execution, and resulting scores and metrics are stored).

• Learning progress data (e.g. completion and performance visible to instructors and students within the platform).

• Instructor-visible analytics (e.g. class and student-level progress, leaderboards, and related reporting).

• AI-assisted learning interactions (e.g. chat and hint usage; conversation data may be stored only for service delivery and is not used to train AI models). These records remain under institutional ownership at all times. AnalytIQ does not own student education records.

4. Institutional Control of Data

Educational institutions retain control over:

• What data is shared with AnalytIQ (e.g. through use of the platform, LTI integration, or roster sync).

• How long data is retained (institutions may request deletion or return in accordance with the Data Processing Agreement and Data Retention and Deletion Policy).

• Who may access student records (access within the platform is governed by role and institutional use; institutions control their own LMS and user management).

• Decisions regarding grading and academic policies (AnalytIQ is not the system of record for official grades; the institution's LMS is when LTI grade passback is used). AnalytIQ acts solely as a data processor and does not make independent decisions about the use or disclosure of education records beyond what is necessary to operate the platform as instructed.

5. Data Use Limitations

AnalytIQ limits the use of education records to:

• Delivering the educational services requested by institutions (e.g. course and class management, exercises, progress tracking).

• Providing AI-assisted learning functionality (e.g. chat, hints) only for service delivery.

• Supporting class management and student progress tracking (e.g. enrollments, progress views, leaderboards).

• Maintaining platform security and reliability (e.g. authentication, access control, logging, audit trails).

AnalytIQ does not:

• Use student data for advertising.

• Sell student information.

• Use student data to train artificial intelligence models.

6. Student Rights Under FERPA

FERPA grants students rights to, among other things:

- Access their education records.
- Request correction of inaccurate records.
- Control disclosure of their information (subject to FERPA's exceptions).

Because AnalytIQ operates under institutional authority as a school official:

- Students should submit FERPA-related requests (e.g. access, correction, disclosure complaints) directly to their educational institution.
- AnalytIQ will assist institutions in fulfilling such requests to the extent required by contract or law (e.g. by providing or deleting data as instructed by the institution).

7. Data Security Safeguards

AnalytIQ maintains administrative, technical, and physical safeguards to protect education

records, including:

- Encryption in transit (all access to the platform over HTTPS/TLS).
- Encryption at rest (via the database and storage infrastructure; sensitive configuration such as SSO secrets is additionally encrypted by the application).
- Role-based access controls (students, instructors, and administrators can access only the data appropriate for their role and permissions).
- Secure cloud infrastructure (hosting and data storage are provided by trusted providers with appropriate security controls).
- Authentication and session protections (e.g. secure password hashing, token-based sessions with expiration, optional SSO and LTI with validated state and signatures).
- Monitoring and audit logging (security-relevant and compliance-relevant events are logged; an audit trail supports accountability and incident response). These safeguards are designed to meet industry standards for educational technology platforms and to support FERPA compliance.

8. Data Retention and Deletion

AnalytIQ retains education records only as long as necessary to provide the services and to meet legal or contractual obligations, as set out in the Data Retention and Deletion Policy.

Institutions may request deletion (or return) of education records at any time in accordance with the Data Processing Agreement. AnalytIQ will process such requests promptly and in line with its contractual obligations and technical capabilities.

9. Subprocessors and Third Parties

AnalytIQ uses trusted service providers (subprocessors) that may process education records on its behalf (e.g. database and storage, application hosting, AI processing, email, error monitoring).

All subprocessors:

• Are contractually bound to maintain appropriate security and data protection safeguards consistent with FERPA and AnalytIQ's commitments.

• Process data solely to support the platform and only on AnalytIQ's instructions.

• Are subject to security and confidentiality requirements in their agreements with AnalytIQ. A current list of subprocessors is maintained in the AnalytIQ Subprocessor List, which is updated periodically.

10. Contact for FERPA Matters

Institutions with FERPA-related questions or requests may contact AnalytIQ through the contact information provided on the AnalytIQ website (e.g. www.myanalytiq.org) or at techsupport@myanalytiq.org or aelharouchi@myanalytiq.org.

AnalytIQ Data Flow Diagram

AnalytIQ Data Flow Diagram This describes how data moves through AnalytIQ. It can be used as a written diagram or converted into a visual chart. A. Core User and Learning Data Flow

1. User creates account or signs in

a. Data: Name, email, and password (or SSO/LMS login). b. Flow: User enters details in the browser → AnalytIQ checks them and creates or finds the account in its database → AnalytIQ sends back a login token (JWT). The browser keeps the token and sends it with later requests. Account data is stored in AnalytIQ's own database (PostgreSQL), not in a separate "Supabase Auth" service.

2. User enrolls in a class and does activities

a. Data: Class enrollment (e.g. class code), which exercises were done, scores, time spent, and completion. b. Flow: User enrolls and does exercises in the browser → browser sends enrollment and progress to AnalytIQ → AnalytIQ saves them in its database. Exercise code is graded in the browser; only the score, accuracy, and time are sent to AnalytIQ and stored. Chapter files (e.g. PDFs) are stored in cloud storage and loaded when the user opens a chapter.

3. Instructor views student progress inside AnalytIQ

a. Data: Progress and results for students in that instructor's classes. b. Flow: Instructor opens the progress/analytics view → their browser asks AnalytIQ for that data → AnalytIQ reads from its database and sends it back → instructor sees it in the interface. The instructor never talks to the database directly; everything goes through AnalytIQ. B. AI Feature Data Flow

1. User sends an AI request (chat or hint)

a. Data: The user's message and, if needed, a bit of context (e.g. current exercise). b. Flow: User types in the chat/hint box → browser sends the message to AnalytIQ.

2. AnalytIQ calls Gemini

a. Data: The same message and context (within size and length limits). b. Flow: AnalytIQ sends the request to Google's Gemini API over a secure connection.

3. User gets the AI reply

a. Data: The AI's reply (text or code suggestion). b. Flow: Gemini sends the reply to AnalytIQ → AnalytIQ sends it to the browser → user sees it in the chat.

Safeguard: These messages are only used to answer the user. They are not used to train AI models. There are also limits on how many requests a user can send. C. Analytics Data Flow a. What happens today: The app does not send product analytics (e.g. clicks, sessions) to Supabase Analytics or PostHog. A product analytics tool (e.g. PostHog) may be added later, only for improving the product and not for ads, and only if allowed by policy. b. Operational data: The app writes logs (e.g. errors, who did what) for running and fixing the service. Some errors may be sent to a monitoring tool (e.g. Sentry) with personal data removed. Important actions (e.g. account or content changes) are also recorded in an audit log in the database. c. Future flow: AnalytIQ application to PostHog D. Email Data Flow a. When: When the system needs to send an email (e.g. password reset, important notice). b. Data: Recipient's email address and the email content. c. Flow: AnalytIQ prepares the email → sends it to an email provider (e.g. Resend) → the provider delivers it to the user's inbox. E. LMS and Grade Record Flow a. LMS link: Schools can connect AnalytIQ to their LMS (e.g. Canvas) using LTI. Users can open AnalytIQ from the LMS and sign in through the school. b. Roster: AnalytIQ can pull the class list from the LMS so enrollments stay in sync. c. Grades: If the school turns on "grade passback," AnalytIQ can send exercise scores to the LMS gradebook. d. Who keeps the official grade: The school's LMS keeps the official grades. AnalytIQ is only a source of scores that can be sent to the LMS when the school has set that up. Security Controls (All Flows)

- HTTPS: All traffic between the browser and AnalytIQ is encrypted.
- Encryption at rest: Data in the database and file storage is encrypted by the cloud provider.
- Role-based access: Users only see what their role allows (e.g. students see their own work, instructors see their classes).
- Strong sign-in: Passwords are hashed; sessions use secure tokens; optional SSO and LMS login are supported.
- Safe coding: The app uses safe database queries and security headers to reduce common web risks.
- Infrastructure: Servers and databases are run by trusted cloud providers (e.g. Supabase, Render) with their own security controls. Summary Diagram (Text) User's browser ■ ■ HTTPS (all requests) ▼ AnalytIQ (Flask app)
- Checks who you are (JWT)
- Saves/loads data
- Calls AI when you chat
- Sends emails when needed

■ ■■■■■■■■■■■■■■■■■ PostgreSQL (users, classes, progress, audit log) ■■■■■■■■■■■■■■■■■ Supabase Storage (chapter files) ■■■■■■■■■■■■■■■■■ Google Gemini (AI replies) ■■■■■■■■■■■■■■■■■ Email provider (e.g. Resend) ■■■■■■■■■■■■■■■■■ LMS (optional: roster +

grades)

AnalytIQ Security Overview Whitepaper

AnalytIQ Security Overview Whitepaper

Last Updated: Feb 20, 2026

Company: AnalytIQ Platform Type: Educational Learning Platform (EdTech) Hosting Region: United States (AWS US West via Supabase)

1. Security Philosophy

AnalytIQ is built with a security-first approach focused on protecting student and institutional data.

We follow three principles:

- Data minimization — We collect and keep only what is needed to run the service.
- Least privilege — Users and systems can access only what their role or function

requires.

- Secure by design — Security is built into the platform and infrastructure, not added on

later. AnalytIQ acts as a data processor for educational institutions. We do not use student data for marketing, advertising, or selling.

2. Infrastructure Security

Cloud hosting

- The AnalytIQ application (web server and optional background workers) is hosted in the

United States using Render (or equivalent) for compute. The application is served over HTTPS and runs behind standard cloud security and availability controls.

- Database and file storage are provided by Supabase (managed PostgreSQL and

S3-compatible object storage), also in the United States. Supabase runs on cloud infrastructure (e.g. AWS) and provides physical and logical security, network controls, redundancy, and availability monitoring for the data layer.

- Caching and rate-limiting use Redis (e.g. Render Redis or similar) for session state,

rate-limit counters, and, when SSO is enabled, OAuth2 state. Redis is used only for transient or non-sensitive operational data; no persistent personal data is stored there beyond what is strictly necessary for the SSO flow.

Data storage All core platform data is stored in an encrypted PostgreSQL database managed through

Supabase. This includes:

- User account information (e.g. identifiers, email, role, authentication-related data)
- Educational records and progress data (enrollments, exercise progress, scores,

completion)

- Course materials and class metadata (classes, chapters, exercises, resources)
- Conversation and AI-related data where needed for service continuity (e.g. chat history)

• Audit log entries for security and compliance

File assets (e.g. chapter PDFs, imported content) are stored in Supabase Storage with encryption at rest provided by the storage provider. Authentication is handled by AnalytIQ itself (JWT issuance and validation). Tokens are not stored server-side except for short-lived, purpose-limited tokens (e.g. password reset) in the database. Session tokens (JWTs) are held in the client (e.g. browser localStorage) and sent with each request; the server validates them and enforces role-based access.

3. Data Encryption

Encryption in transit:

All data between users and AnalytIQ is protected with HTTPS (TLS). This applies to:

- Login and authentication (including SSO and LTI)
- Learning data (enrollments, progress, scores, content)
- AI feature requests and responses
- LMS integrations (LTI launch, roster sync, grade passback)

Outbound calls from AnalytIQ to third-party services (e.g. Google Gemini, email provider) use TLS.

Encryption at rest:

- Stored data is encrypted at rest using the encryption mechanisms of the underlying

providers (Supabase for database and object storage, and the hosting provider for any transient or log data). AnalytIQ does not implement its own full-disk or database encryption; it relies on these infrastructure-level controls. Sensitive configuration (e.g. SSO client secrets) is encrypted with a symmetric key derived from the application secret before being stored in the database.

4. Access Control and Authentication

Role-based access control:

AnalytIQ enforces strict role-based permissions:

- Students can access only their own profile, enrollments, progress, and the classes they

are enrolled in.

- Professors (instructors) can access only the classes they own and the associated

students, chapters, exercises, progress, and analytics.

- Tenant administrators can manage users, settings, and SSO configuration only within

their tenant.

- Superadministrators have platform-wide administrative access, subject to the same

authentication and audit controls. Every API request is authenticated (JWT) and authorized according to role and resource ownership; there is no direct database access from the client. Authentication security

- Passwords are hashed with bcrypt (configurable work factor) and never stored in plain

text. Legacy hash formats are supported only for migration and are rehashed on successful login where applicable.

• Sessions are token-based (JWT). Tokens are signed with a server-held secret, include

role and (where applicable) tenant scope, and have a configurable expiration. There are no server-side session stores; revocation is achieved by token expiry or by disabling the account.

• Session expiration is enforced on every request; expired or invalid tokens result in 401

and, in the client, redirect to login and clearing of stored credentials.

• Optional SSO (per-tenant) uses OAuth2/OpenID Connect with state stored in Redis to

prevent CSRF; JWTs are issued by AnalytIQ after successful IdP authentication. LTI 1.3 uses OIDC and signed id_tokens from the LMS; AnalytIQ validates signatures and issues its own JWT for the session.

• Administrative access is restricted to users with the appropriate roles (e.g. tenant_admin,

superadmin) and is protected by the same authentication and RBAC mechanisms; administrative actions are logged in the audit log. .

5. Application Security

Secure development practices

AnalytIQ applies secure coding practices, including:

• Input validation and length limits for user-supplied data (names, emails, passwords,

class and chapter metadata, exercise content, AI messages), with centralized limits defined in configuration.

• Parameterized database queries to prevent SQL injection; the repository layer uses

bound parameters for all SQL.

• Protection against common web risks via security headers (e.g.

X-Content-Type-Options, X-Frame-Options, Referrer-Policy, and in production HSTS), optional CSRF protection when enabled, and no reliance on client-supplied data for authorization beyond what is validated in the token and database.

• Rate limiting (e.g. Flask-Limiter with Redis or in-memory backend) on authentication and

sensitive endpoints (signup, sign-in, password reset, AI chat) to reduce abuse and brute-force risk.

• Dependency and code review as part of the development and release process;

dependencies are updated and security-relevant changes are reviewed before deployment. Monitoring and logging

• The platform uses structured logging (e.g. JSON format with request identifiers) for

operational and security-relevant events. Logs are retained according to the application and hosting provider’s retention and access policies (e.g. Render); they are not governed by Supabase retention policies alone.

• Audit logging is implemented in the application: designated mutations (e.g. user creation,

password reset, class and enrollment changes, chapter and exercise changes) are written to an audit_log table in the database. Entries are committed in the same transaction as the business operation to support accountability and compliance.

• Error and performance monitoring may use a third-party service (e.g. Sentry) with PII

filtering so that sensitive data is not sent to the monitoring provider. Logs and audit data are used for incident detection, investigation, and response.

6. AI Security and Data Protection

AnalytIQ provides AI-powered educational assistance (e.g. chat, hints, exercise generation) using the Google Gemini API.

Safeguards:

• AI prompts and context are sent to Gemini only to generate responses for the user or for content generation (e.g. exercises from chapter text). They are not used to train AnalytIQ or third-party models.

• Conversation history may be stored in the platform's database to support multi-turn chat and continuity within the session; such storage is for service delivery only and is subject to the same access control and retention policies as other user data.

• Rate and size limits apply to AI chat (e.g. per-user per minute and per hour, and caps on message size and conversation length) to limit abuse and resource consumption.

• Grading of student code is performed server-side using sandboxed subprocess execution with AI-generated unit test cases. The execution environment enforces memory limits, import controls, and process isolation. AI generates the test cases during exercise creation (under instructor review); AI does not make real-time grading decisions during student submission. Scores and related metrics are stored on the backend.

AI features are implemented as educational assistance tools under human oversight; they do not replace instructor judgment or institutional grading policy.

7. Data Privacy and FERPA Compliance

AnalytIQ operates as a school official with a legitimate educational interest under FERPA when processing education records on behalf of institutions.

The platform:

• Processes student data only on behalf of institutions and in accordance with contractual and documented instructions.

• Does not sell or share student data for marketing or commercial purposes.

• Uses data solely for educational functionality, platform operations, and compliance (e.g. audit logs).

• Does not act as the official system of record for grades; the institution's LMS retains that role when LTI is used, and grade passback (when enabled) is under institutional control.

8. Incident Detection and Response

AnalytIQ maintains procedures to identify and respond to security incidents, including:

• Ongoing monitoring of application and infrastructure (logs, errors, and, where deployed, error-tracking alerts).

• Structured logging and audit trails to support investigation of suspicious or anomalous activity.

• Incident investigation using available logs, audit records, and provider-side information as appropriate.

• Notification to affected institutions where required by law or contract (e.g. personal data breach notification).

9. Subprocessor Security

AnalytIQ uses a limited set of subprocessors necessary to operate the platform, including:

• Supabase — managed PostgreSQL database and object storage (United States).

• Render (or equivalent) — application hosting and, where used, Redis for caching and rate limiting.

• Google Gemini API — AI processing for chat, hints, and content generation.

• Email provider (e.g. Resend) — transactional email (e.g. password reset, notifications).

• Error and performance monitoring (e.g. Sentry) — with contractual and technical measures to limit exposure of personal data (e.g. PII filtering). Product analytics (e.g. PostHog) are not currently in use; any future use would be subject to legal and policy review (e.g. FERPA) and would be for product improvement only, not advertising. Subprocessors are selected with regard to security and data protection and are contractually required (where applicable) to maintain appropriate safeguards. AnalytIQ remains responsible for subprocessor compliance with the same standards reflected in this whitepaper and in the Data Processing Agreement.

10. Data Retention and Deletion

• Personal data is retained only as long as necessary to provide the service and to meet legal or contractual obligations (e.g. audit, legal hold).

• Upon account deletion or upon written request by the institution, AnalytIQ will, at the controller's choice and subject to applicable law, permanently delete the relevant personal data from active systems or securely return it in a structured format. Any retention beyond that is only where required by law.

• Educational institutions remain responsible for records they retain in their own systems (e.g. LMS); AnalytIQ does not control those retention periods.

11. Compliance Alignment

AnalytIQ's security practices are designed to align with:

• FERPA data protection requirements for education records.

• U.S. state privacy laws applicable to the services and data processed.

• Transparency and accountability expectations for international data transfer and processor relationships, as reflected in contracts and documentation.

• Industry best practices for cloud-hosted applications, including encryption, access control, logging, and secure development.

12. Contact for Security Inquiries

Security questions may be directed to AnalytIQ through the contact information provided on the platform website (e.g. www.myanalytiq.org).

AnalytIQ Incident Response Policy

AnalytIQ Incident Response Policy

Last Updated: Feb 20, 2026

Company: AnalytIQ Scope: All systems and data processed by the AnalytIQ platform

1. Purpose

This policy defines how AnalytIQ detects, responds to, and mitigates security incidents that involve unauthorized access, disclosure, alteration, or destruction of data. The goals are to contain impact quickly, protect educational records, and communicate in a timely way with affected institutions and, where required, with regulators or data subjects.

2. Definition of a Security Incident

A security incident is any confirmed or reasonably suspected event that compromises the confidentiality, integrity, or availability of data processed by AnalytIQ.

Examples include:

- Unauthorized access to user accounts or administrative interfaces
- Unauthorized disclosure or exposure of personal or educational records
- System intrusion, malware, or other malicious activity
- Loss or theft of sensitive data (e.g. through breach or misconfiguration)
- Significant service disruption caused by a cyber incident (e.g. denial of service, ransomware)
- Misuse of the platform (e.g. abuse of AI or authentication endpoints) that poses a material risk to users or institutions

3. Incident Detection

AnalytIQ uses several detection mechanisms to identify potential incidents:

- Application and security logging — Structured logs (e.g. JSON with request identifiers) capture requests, errors, and security-relevant events. Logs are retained according to the application and hosting provider's retention policy and are used for review and investigation.
- Audit trail — Designated user and data mutations (e.g. account creation, password reset, class and enrollment changes, content changes) are recorded in an audit log in the database, supporting accountability and incident investigation.
- Error and performance monitoring — Where deployed, a third-party monitoring service (e.g. Sentry) is used to alert on application errors and failures. PII is filtered before data is sent to the monitor.
- Infrastructure and provider notifications — Alerts or notifications from hosting,

database, and storage providers (e.g. Render, Supabase) regarding availability, security, or abuse are treated as potential incident indicators.

• User and institution reports — Reports received via the platform's contact channel or from institutional contacts are logged and assessed as potential incidents. All detected anomalies and reported events are reviewed promptly to determine whether they qualify as a security incident and require formal response.

4. Incident Response Process

AnalytIQ follows a structured response process in four stages.

Stage 1: Identification

• The event is logged and classified (e.g. type, severity, affected scope).

• An initial assessment is performed to determine scope and severity.

• Affected systems and data types are identified (e.g. which environment, which data stores, whether personal or educational records are involved).

Stage 2: Containment

• Immediate steps are taken to limit impact. Depending on the incident, these may include:

• Restricting access — Use of platform-level controls such as IP blocklisting (where supported) to block abusive or malicious sources.

• Revoking or limiting compromised access — Disabling affected user accounts, forcing password reset, or restricting administrative access as appropriate.

• Blocking or throttling malicious activity — Leveraging rate limiting and, where applicable, provider-level controls to reduce ongoing abuse.

• Where the incident affects infrastructure or third-party services, containment may also involve coordination with the hosting or database provider (e.g. isolation of affected systems or accounts at the provider level).

Stage 3: Investigation and Remediation

A thorough investigation is conducted to determine:

• Root cause of the incident

• Extent of data exposure or impact

• Corrective actions required

Remediation may include:

• Applying security patches or dependency updates

• Access control and configuration changes (e.g. credential rotation, role or permission adjustments)

• Infrastructure or application configuration changes to close gaps or harden controls
• Process or policy updates where the incident reveals procedural weaknesses

Stage 4: Recovery

• Affected systems are restored to normal operation once it is safe to do so.
• Additional monitoring may be put in place for a period to ensure stability and to detect recurrence.
• Post-incident review (see Section 8) is performed to capture lessons learned and improve future response.

5. Notification Procedures

If a confirmed incident involves unauthorized access to or disclosure of personal or educational

data:

AnalytIQ will notify affected institutions without undue delay after the incident is confirmed and key facts are known. Notification will include, to the extent known and legally permissible:

• A description of the incident
• The types of data involved
• Steps taken to contain the incident and mitigate harm
• Recommended actions for the institution (e.g. user communication, credential resets)

AnalytIQ will cooperate with institutions in meeting any legal or regulatory reporting obligations (e.g. breach notification to authorities or data subjects) that apply to the controller or to AnalytIQ as processor. Notification procedures may be updated to reflect specific contractual or regulatory requirements (e.g. DPA, FERPA, state breach laws).

6. Responsibilities

AnalytIQ responsibilities:

• Maintain logging, audit, and monitoring capabilities used for incident detection
• Investigate incidents promptly and in accordance with this policy
• Implement containment and remediation measures as appropriate
• Notify institutions when required by law or contract
• Document incident response actions and outcomes (see Section 7)

Institutional responsibilities:

• Educational institutions (controllers) are responsible for:
• Managing user credentials and access within their own environment (e.g. LMS, SSO)
• Maintaining the security of their LMS and related systems that integrate with AnalytIQ
• Communicating with students and staff when necessary (e.g. after a breach notification

from AnalytIQ)

• Meeting their own legal obligations (e.g. breach notification to regulators or data subjects)

7. Incident Documentation

AnalytIQ maintains records of security incidents, including:

• Timeline of events (detection, assessment, containment, remediation, recovery)
• Impact assessment (systems and data affected, severity)
• Remediation steps taken
• Lessons learned and any follow-up actions

These records are retained for internal security review, compliance, and continuous improvement. Retention is consistent with legal and contractual requirements and with AnalytIQ's data retention practices.

8. Continuous Improvement

After each incident, AnalytIQ evaluates:

• Response effectiveness (speed, appropriateness of containment and remediation)
• Gaps in security controls or detection capabilities
• Preventive improvements (technical, procedural, or policy)

Security practices, runbooks, and this policy are updated as needed to reflect lessons learned and changes in technology, infrastructure, or risk.

9. Policy Review

This policy is reviewed periodically and updated as necessary to reflect:

• Changes in technology or infrastructure
• Legal or regulatory requirements
• Contractual obligations (e.g. DPA, customer agreements)

- Lessons learnedfrom incidentsorexercises

A nalytIQDataRetentionandDeletion Policy

AnalytIQ Data Retention and Deletion Policy

Last Updated: Feb 20, 2026

Company: AnalytIQ Scope: All personal data and educational records processed by the AnalytIQ platform.

1. Purpose

This policy describes how AnalytIQ retains, manages, and deletes personal data and educational records processed on behalf of users and educational institutions. The goal is to ensure that data is retained only as long as necessary to provide the service and to meet legal or contractual obligations, and is securely deleted or returned when no longer required.

2. Data Categories Covered

This policy applies to data processed by AnalytIQ, including:

• Account information — Name, email, role, authentication-related data (e.g. password hash or SSO/LTI identifiers), account status, and (where applicable) tenant association.

• Educational records — Class enrollments, progress data (e.g. exercise scores, completion, time spent), and related identifiers.

• Course and content data — Classes, chapters, exercises, and resources created by instructors; metadata is stored in the database; files (e.g. chapter PDFs) are stored in object storage.

• AI-related data — Conversation and message content used to provide chat and continuity; AI digest or similar context where stored for the service. Such data is used only for service delivery and is not used to train models.

• System and audit data — Application logs (e.g. request and error logs), audit log entries (e.g. user and resource mutations), and, where used, error-monitoring data (with PII reduced or removed).

3. Data Retention Principles

AnalytIQ applies the following principles:

• Data is retained only for legitimate educational and operational purposes (e.g. service delivery, security, compliance).

• Retention periods are minimized where possible, subject to law and contract.

• Where AnalytIQ is used under an institutional agreement, educational institutions retain primary responsibility for deciding how long education records must be kept for academic or regulatory purposes; AnalytIQ processes such data only in accordance with instructions and this policy.

• Data is not retained for marketing or commercial profiling.

4. Account and Educational Data

Retention Active accounts

• For active users, personal and educational data is retained for the duration of the account's use of the service.

• This allows users and instructors to access current and historical progress, course records, and content as intended by the service. Account deletion requests

• When a user requests account deletion (or when deletion is performed at the request of

an institution or under contract):

• Personal data associated with the account is permanently removed from active systems without undue delay, in line with the platform's deletion capabilities (e.g. user record and, where applicable, related profile, settings, preferences, progress, and conversation data).

• Content that the user created (e.g. classes, chapters, exercises) may be retained, deleted, or transferred according to product behavior and contractual or institutional instructions.

• Educational institutions may retain their own copies of education records in their systems (e.g. LMS) independently of AnalytIQ; this policy does not govern institutional retention. Institutional data control

• Where AnalytIQ is used under an institutional or tenant agreement, the institution (or tenant) determines how long educational records must be retained to meet academic or regulatory requirements.

• AnalytIQ processes such data only under institutional direction and in accordance with the Data Processing Agreement and this policy.

5. AI Data Retention

• AI prompts and responses are used only to deliver the AI feature (e.g. chat, hints).

• Conversation and message data may be stored in the platform's database to support multi-turn chat and continuity within the session; such storage is for service delivery only.

• AnalytIQ does not use AI prompt or response data to train its own or third-party AI models.

• Retention of AI-related data follows the same account and deletion rules (e.g. removed or anonymized when the account or related data is deleted) unless a specific exception under Section 10 applies.

6. System Logs and Analytics Data

• Operational logs (e.g. application and request logs) are retained according to the application and hosting provider's retention and access policies (e.g. Render). They are not governed by database or storage retention alone.

• Audit log entries (e.g. security and compliance-relevant events) are stored in the database and retained for a period necessary for security review, compliance, and incident response, consistent with law and contract.

• Error and performance monitoring data (e.g. Sentry), where used, is retained per the monitoring provider's terms; PII is filtered before data is sent.

• These logs and monitoring data are used only for security monitoring, performance analysis, and troubleshooting. They do not contain full educational records, though they may include identifiers (e.g. user ID, request ID) where necessary for investigation.

• Product analytics (e.g. PostHog) are not currently in use; any future use would be subject to legal and policy review and would not include retention for advertising or profiling.

7. Backup Data Retention

• Backups of the database and, where applicable, configuration or other backed-up data are managed by the infrastructure and database provider (e.g. Supabase) or the hosting provider.

• Backup retention follows the provider’s schedules and policies. Backups may be overwritten or deleted automatically according to those schedules.

• AnalytIQ does not control the exact retention period of provider-managed backups; upon contract termination or deletion requests, active data is deleted or returned as set out in this policy and the DPA, and backup data will be purged through the provider’s normal retention and overwrite cycles..

8. Data Deletion Upon Contract

Termination When an institution or tenant terminates its use of the service (or when a data deletion

obligation is triggered under the contract):

• AnalytIQ will, upon written request from the institution (or as agreed in the contract):

• Delete the institution’s (or tenant’s) data from active systems, or

• Return the data in a reasonable, machine-readable format where technically feasible and requested. Backup data will be removed through the provider’s normal automated retention and overwrite cycles; AnalytIQ does not retain backups beyond those cycles for the purpose of retaining terminated-institution data.

9. Secure Deletion Practices

When data is deleted under this policy, AnalytIQ applies secure deletion practices appropriate to

the system, including:

• Removal of data from active databases (e.g. deletion or cascade of related rows so that the data is no longer accessible through normal application or database access).

• Revocation of access — Affected user accounts are disabled or removed so that credentials (e.g. JWT) no longer grant access to that data.

• Deletion of associated storage objects — Where files or other objects (e.g. chapter files in object storage) are linked to deleted accounts or content, such objects are deleted from active storage where the product and operational design support it. These practices ensure that deleted data cannot be accessed through normal system operations. AnalytIQ does not commit to cryptographic erasure of storage media; reliance is on deletion at the application and storage layer and on provider retention and overwrite for backups.

10. Exceptions to Deletion

AnalytIQ may retain limited data when required by law or for legitimate purposes, such as:

• Legal hold or regulatory obligation

• Security or fraud investigations

• Dispute resolution or enforcement of agreements

• Compliance with applicable retention obligations

Such data is retained only as long as necessary for the stated purpose and is subject to appropriate access and security controls. Where retention is based on legal obligation, AnalytIQ will not use the data for other

purposes and will delete or anonymize it when the obligation no longer applies, unless law requires otherwise.

11. Policy Review

This policy is reviewed periodically and updated as necessary to reflect:

- Operational changes (e.g. new data categories, new features, new providers)
- Legal or regulatory requirements
- Contractual obligations (e.g. DPA, customer agreements)

AnalytIQ Security Questionnaire

Response Template

AnalytIQ Security Questionnaire

Response Template Company: AnalytIQ Platform Type: Educational Learning Platform (EdTech) Hosting Environment: United States — application hosting via Render; database and object storage via Supabase Primary Data Type: Educational records and account data

SECTION 1. General Information

Q: Describe your service.

AnalytIQ is an educational learning platform that provides course management, practice exercises (including coding exercises), AI-assisted learning support (e.g. chat and hints), progress tracking, and integration with institutional learning management systems (LTI 1.3). It processes data to deliver these services on behalf of users and institutions.

Q: What types of data do you process?

AnalytIQ processes account information (e.g. name, email, role, authentication data), educational records (enrollments, progress, scores, completion), course materials and metadata (classes, chapters, exercises, resources), AI conversation data where needed for service continuity, and limited technical data (e.g. logs, audit records) for security and operations. It does not process data for advertising or product analytics beyond what is needed for the service.

Q: Do you sell personal data?

No. AnalytIQ does not sell personal data.

Q: Do you use data for advertising?

No. AnalytIQ does not use personal data for advertising or marketing.

SECTION 2. Data Hosting and

Infrastructure

Q: Where is data hosted?

All data is hosted in the United States. The application runs on Render (or equivalent). The database (PostgreSQL) and object storage (e.g. chapter files) are provided by Supabase. Supabase may use underlying cloud infrastructure (e.g. AWS); AnalytIQ does not operate its own data centers.

Q: Is data encrypted at rest?

Yes. Data stored in the database and in object storage is encrypted at rest using the encryption mechanisms provided by the infrastructure providers (e.g. Supabase). Sensitive configuration (e.g. SSO client secrets) is additionally encrypted by the application before storage.

Q: Is data encrypted in transit?

Yes. All data transmission between users and AnalytIQ, and between AnalytIQ and third-party services, uses HTTPS (TLS).

Q: Do you use third-party cloud providers?

Yes. AnalytIQ uses third-party providers including Supabase (database and object storage), Render (application hosting and, where used, Redis for caching and rate limiting), Google Gemini API (AI features), an email provider (e.g. Resend) for transactional email, and optionally error-monitoring services (e.g. Sentry)

with PII filtering.

SECTION 3. Access Control

Q: How is access to data restricted?

AnalytIQ uses role-based access control (RBAC). Users can access only the data required for their role: students see their own profile, enrollments, and progress; instructors see only their classes and related students; tenant and platform administrators have scoped access. Every API request is authenticated (JWT) and authorized by the backend; there is no direct client access to the database.

Q: Do administrators have unrestricted access?

No. Administrative access is limited to users with tenant_admin or superadmin roles and is enforced by the same authentication and RBAC. Administrators can access only what their role and scope allow; actions are logged in an audit trail.

Q: Are passwords securely stored?

Yes. Passwords are stored using bcrypt hashing (configurable work factor). Plain-text passwords are not stored. Legacy hash formats are supported only for migration and are rehashed on successful login where applicable.

SECTION 4. Authentication and Account

Security

Q: How are user sessions protected?

AnalytIQ uses token-based authentication (JWT). Tokens are signed with a server-held secret, include role and (where applicable) tenant scope, and have a configurable expiration. Expired or invalid tokens are rejected; the client clears stored credentials and redirects to login on 401. Optional per-tenant SSO (OAuth2/OIDC) and LTI 1.3 use state and signature validation to protect launch and callback flows.

Q: Do you support multi-factor authentication (MFA)?

Multi-factor authentication is not currently implemented at the application level. Institutions using LTI 1.3 or SSO benefit from their identity provider’s MFA capabilities. Application-level MFA is on the product roadmap.

SECTION 5. AI Data Handling

Q: Does AI training use customer data?

No. AnalytIQ does not use student or customer data to train AI models.

Q: What AI provider is used?

Google Gemini API is used to provide AI assistance (e.g. chat, hints, exercise generation). Requests and responses are used only to deliver the feature and are not used for model training.

Q: Is AI data retained?

AI prompts and responses are used only for service delivery. Conversation and message data may be stored in the platform’s database to support multi-turn chat and continuity; such storage is not for training and is subject to the same retention and deletion policy as other user data (e.g. removed or anonymized on account deletion where applicable).

SECTION 6. Monitoring and Logging

Q: Do you maintain system logs?

Yes. AnalytIQ maintains structured application logs (e.g. request identifiers, errors, security-relevant events) and an audit log in the database for designated user and data mutations. Where deployed, error and performance monitoring (e.g. Sentry) is used with PII filtered before data is sent. Logs are used for security monitoring, troubleshooting, and compliance.

Q: How long are logs retained?

Application logs are retained according to the application and hosting provider's retention and access policies (e.g. Render). Audit log entries in the database are retained for a period necessary for security, compliance, and incident response, consistent with legal and contractual requirements. Retention is not governed solely by Supabase infrastructure defaults.

SECTION 7. Incident Response

Q: Do you have an incident response plan?

Yes. AnalytIQ maintains an Incident Response Policy that defines how security incidents are detected, assessed, contained, investigated, remediated, and recovered from, and how institutions are notified when required.

Q: How quickly do you notify customers of breaches?

Affected institutions are notified without undue delay after a confirmed incident involving unauthorized access to or disclosure of personal or educational data, and once key facts are known. Notification includes a description of the incident, types of data involved, steps taken, and recommended actions, and we cooperate with institutions in meeting legal reporting obligations.

SECTION 8. Data Retention

Q: How long is personal data retained?

Personal data is retained only as long as necessary to provide the service and to meet legal or contractual obligations (e.g. audit, legal hold). Retention is described in AnalytIQ's Data Retention and Deletion Policy.

Q: Can customers request deletion?

Yes. Users and institutions may request deletion of their data. Upon valid request (or upon contract termination), AnalytIQ will delete the relevant data from active systems or return it in a reasonable format, as set out in the Data Processing Agreement and the Data Retention and Deletion Policy.

SECTION 9. Compliance

Q: Are you FERPA compliant?

AnalytIQ operates as a school official with a legitimate educational interest under FERPA when processing education records on behalf of institutions. Student data is processed only for the contracted educational services and is not used for marketing, resale, or AI model training.

Q: Are you GDPR compliant?

AnalytIQ operates as a data processor and processes personal data only on documented instructions from the controller (institution or user, as applicable). Data processing agreements and technical and organizational measures are designed to support controller compliance with applicable law, including GDPR where it applies.

Q: Do you have SOC 2 certification?

AnalytIQ does not currently hold SOC 2 certification but follows industry-standard security practices (e.g. encryption, access control, logging, incident response, retention and deletion) as described in its security documentation and policies.

SECTION 10. Subprocessors

Q: Do you use subprocessors?

Yes. AnalytIQ uses a limited set of subprocessors, including Supabase (database and object storage), Render (application hosting and Redis), Google Gemini API (AI processing), an email provider (e.g. Resend), and optionally error-monitoring (e.g. Sentry). Product analytics (e.g. PostHog) are not currently in use; any future use would be subject to legal and policy review. Subprocessors are contractually required to maintain appropriate security and data protection safeguards.

SECTION 11. Data Backup and Recovery

Q: Do you maintain backups?

Backups of the database and, where applicable, other backed-up data are maintained by the infrastructure and database provider (e.g. Supabase) or the hosting provider. Such backups support disaster recovery and service continuity.

Q: How is backup data protected?

Backup data is stored within the hosting provider's infrastructure and is protected by the provider's controls, including encryption at rest and access controls. Backup retention and overwrite follow the provider's schedules; upon deletion requests or contract termination, active data is deleted or returned, and backup data is purged through normal provider retention cycles.

SECTION 12. Physical Security

Q: Do you maintain physical data centers?

No. AnalytIQ does not operate physical data centers. Physical security is provided by its

infrastructure providers (e.g. Render for application hosting, AWS through Supabase for database and storage). Those providers operate in the United States and maintain physical and environmental security for their facilities.

AnalytIQ Vendor Readiness Packet

AnalytIQ Vendor Readiness Packet

Company: AnalytIQ Platform Type: Educational Learning Platform (EdTech) Hosting Region: United States (AWS US West via Supabase) Primary Data Type: Educational records and account data

1. Company Overview

AnalytIQ is an educational platform that supports teaching and learning in data analytics and coding.

The platform provides:

• Course and class management — Create classes, manage chapters and exercises, and organize resources.

• Interactive practice exercises — Coding exercises (including AI-generated ones) with in-browser editing and server-side sandboxed grading (subprocess isolation, memory caps, import controls, AI-generated unit tests); scores and progress are stored.

• AI-assisted learning support — Chat and contextual hints powered by the Google Gemini API.

• Learning progress tracking — Per-student and class-level progress, completion, and leaderboards.

• Integration with institutional learning management systems — LTI 1.3 for launch, roster sync (NRPS), and optional grade passback (AGS). AnalytIQ operates as a data processor on behalf of educational institutions and users. It does not own student educational records; the institution's LMS remains the system of record for official grades when LTI is used.

2. Security Overview

Infrastructure security

• Application hosting — The AnalytIQ application (web server and optional background workers) is hosted in the United States using Render (or equivalent), with standard cloud security and availability controls.

• Database and storage — Core platform data is stored in a PostgreSQL database and object storage managed by Supabase (also in the United States). Supabase provides encryption at rest, access controls, backups, and monitoring for the data layer; it may use underlying cloud infrastructure (e.g. AWS).

• Caching and rate limiting — Redis (e.g. via Render) is used for rate limiting, caching, and, when SSO is enabled, OAuth2 state. It is not used for persistent storage of personal data. AnalytIQ does not operate its own data centers; physical and network security are provided by these infrastructure providers. Encryption Encryption in transit All data transmitted between users and the platform, and between the platform and third-party services, is protected using HTTPS (TLS). Encryption at rest Stored data is encrypted at rest using the encryption mechanisms of the infrastructure providers (e.g. Supabase for database and object storage). Sensitive configuration (e.g. SSO client secrets) is additionally encrypted by the application before storage. Access control

AnalytIQ uses role-based access control (RBAC):

• Students can access only their own profile, enrollments, progress, and enrolled classes.

• Instructors (professors) can access only the classes they own and the associated students, chapters, exercises, and analytics.

• Tenant and platform administrators have scoped access (e.g. tenant_admin for their tenant, superadmin for the platform); administrative actions are logged.

• All access is enforced by the backend after authentication; there is no direct client access to the database. Authentication security

• Passwords are stored using bcrypt hashing; plain-text passwords are not stored.

• Sessions use token-based authentication (JWT) with configurable expiration; expired or invalid tokens are rejected.

• Optional per-tenant SSO (OAuth2/OIDC) and LTI 1.3 are supported; state and signatures are validated to protect against misuse.

3. Data Privacy and FERPA Compliance

AnalytIQ operates as a school official with a legitimate educational interest under the U.S. Family Educational Rights and Privacy Act (FERPA) when processing education records on behalf of institutions.

The platform:

• Processes student data only under institutional direction and in accordance with the Data Processing Agreement and documented instructions.

• Does not sell personal information.

• Uses data solely for educational functionality, platform operations, and compliance (e.g. audit logs).

• Does not use student data for AI model training; AI features are for service delivery only.

Official academic records and grades remain under institutional control. When LTI is used, the institution's LMS is the system of record; AnalytIQ can send grades to the LMS via LTI AGS when the institution has enabled that feature.

4. Subprocessors

AnalytIQ uses a limited set of subprocessors:

Hosting and infrastructure

• Supabase — Managed PostgreSQL database and S3-compatible object storage (United States).

• Render — Application hosting and Redis (e.g. for rate limiting, caching, SSO state).

• Supabase may use underlying cloud infrastructure (e.g. AWS); AnalytIQ does not directly contract with AWS for application hosting. Authentication

• Authentication (sign-up, sign-in, password reset, JWT issuance) is handled by AnalytIQ itself; Supabase is not used as an authentication provider for user logins. AI processing

• Google Gemini API — Used for AI-assisted features (chat, hints, exercise generation).

• Prompts and responses are used only for service delivery. Conversation data may be stored in the platform's database for chat continuity; it is not used for model training. Communications

• Transactional email provider (e.g. Resend) — Password reset, security-related notices, and other transactional email. Monitoring

• Error and performance monitoring (e.g. Sentry) — With PII filtered before data is sent.

• Product analytics (e.g. PostHog) are not currently in use; any future use would be for product improvement only, subject to legal and policy review (e.g. FERPA), and would not be used for advertising. Subprocessors are required to maintain appropriate security and data protection safeguards; AnalytIQ remains responsible for subprocessor compliance.

5. Data Flow Overview

Core data flow

• Users submit account information and learning data (enrollments, progress, scores, content) through the platform.

• Data is transmitted over HTTPS and stored in the PostgreSQL database and object storage (Supabase). The application runs on Render and does not store persistent personal data on application servers.

• Instructors access student progress through the platform's web interface; all data access is via the AnalytIQ backend with role-based authorization. AI processing flow

• User messages are sent securely from the AnalytIQ backend to the Google Gemini API to generate responses.

• Conversation data may be stored in the platform's database for multi-turn chat continuity; it is not used for AI training. Rate and size limits apply to AI usage. Analytics and logging flow

• Application and audit logs are retained for security, troubleshooting, and compliance; they are held according to the application and hosting provider's policies and the platform's retention policy.

• Product usage analytics (e.g. feature clicks, sessions) are not currently collected; any future use would be disclosed and subject to legal and policy review. LMS integration flow

• AnalytIQ may integrate with institutional LMS systems via LTI 1.3 (launch, roster sync, optional grade passback).

• Final academic records remain under institutional control; the LMS is the system of record for official grades.

6. Incident Response

AnalytIQ maintains procedures to detect and respond to security incidents, as set out in its

Incident Response Policy. These include:

• Monitoring and alerting — Application logs, audit trail, and (where deployed) error-monitoring alerts.

• Review of anomalies — Including reports from users, institutions, or infrastructure providers.

• Structured response — Identification, containment (e.g. IP blocklisting, account disable, rate limiting), investigation, remediation, and recovery.

• Notification — Affected institutions are notified without undue delay after confirmation of an incident involving personal or educational data, with a description of the incident, data involved, steps taken, and recommended actions.

7. Data Retention and Deletion

Personal and educational data is retained only as long as necessary to provide the service and to meet legal or contractual obligations, in line with the Data Retention and Deletion Policy. When a user deletes their account (or when deletion is requested by an institution or under

contract):

• Personal data associated with the account is permanently removed from active systems without undue delay, consistent with the platform's deletion capabilities.

• Educational institutions retain control over records stored in their own systems (e.g. LMS); AnalytIQ does not govern institutional retention. Backups are maintained by infrastructure providers (e.g. Supabase, Render) for disaster recovery; backup data is encrypted and overwritten or deleted according to the provider's retention cycles. Upon contract termination or deletion requests, active data is deleted or returned, and backup data is purged through normal provider processes.

8. Compliance Alignment

AnalytIQ's security and privacy practices are designed to align with:

• FERPA requirements for educational data protection (school official, legitimate educational interest).

• U.S. state privacy laws (e.g. CCPA) where applicable to the services and data processed.

• International data transfer and transparency expectations, as reflected in contracts and documentation (e.g. DPA, processor obligations).

• Industry best practices for cloud-hosted applications (encryption, access control, logging, incident response, retention and deletion). AnalytIQ does not currently hold SOC 2 or similar certifications but follows the practices described in this packet and in its security and privacy policies.

9. Contact Information

Security or compliance inquiries may be directed to AnalytIQ through the contact information provided on the platform website (e.g. www.myanalytiq.org).

AnalytIQ Subprocessor List

AnalytIQ Subprocessor List

Last Updated: February 20, 2026

This document lists third-party service providers (“Subprocessors”) that AnalytIQ may use to process customer or student data in connection with the AnalytIQ platform. All Subprocessors are contractually required to maintain appropriate security and confidentiality safeguards and may process data only as necessary to provide their services. AnalytIQ remains responsible for Subprocessor compliance with our data protection commitments.

1. Infrastructure and Hosting

Render Services, Inc. (or equivalent):

• Purpose: Application hosting (web server and optional background workers); Redis for rate limiting, caching, and (when SSO is enabled) OAuth2 state storage.

• Data processed: Data in transit through the application; request and error logs; transient Redis data (e.g. rate-limit counters, SSO state tokens). No persistent storage of personal or educational records in Redis beyond short-lived SSO state.

• Hosting location: United States.

Supabase, Inc.

• Purpose: Managed PostgreSQL database and object storage (S3-compatible) for the platform. Used for all persistent application data and stored files (e.g. chapter PDFs, imported content).

• Data processed: User account information; educational records (enrollments, progress, scores); course and class data; chapter and exercise content; uploaded and imported files; audit logs; AI conversation data where stored for service continuity.

• Hosting location: United States (Supabase may use underlying cloud infrastructure such as AWS in the same region). Amazon Web Services (AWS)

• Purpose: Underlying cloud infrastructure used by Supabase for database and storage services. AnalytIQ does not directly contract with AWS for application hosting; the application runs on Render.

• Data processed: Data stored in or transmitted through Supabase-managed services (as above).

• Hosting location: United States.

2. Artificial Intelligence Processing

Google LLC (Gemini API)

• Purpose: AI-powered learning assistance (e.g. chat, hints) and content generation (e.g. exercises from course material).

• Data processed: User prompts and context sent in real time to generate responses. Conversation history may be stored in AnalytIQ’s database for continuity; Google processes only the data sent per request.

• Restrictions: Data sent to Google is used only to generate responses. It is not retained by AnalytIQ or Google for AI model training and is not used for advertising. Rate and size limits apply.

• Processing location: United States (or as applicable for Google Cloud services).

3. Communication Services

Resend (or equivalent transactional email provider)

• Purpose: Sending transactional email (e.g. password reset, security-related notices, administrative notifications).

• Data processed: Recipient email addresses and the content of the emails (e.g. reset links, notices). No sensitive credentials in message body.

• Processing location: United States (or as disclosed by the provider).

4. Monitoring and Error Tracking

Sentry (Functional Software, Inc.)

• Purpose: Error monitoring and performance tracking to support reliability and troubleshooting.

• Data processed: Technical error logs, stack traces, and diagnostic metadata. Personal data and sensitive content are filtered or removed before data is sent to Sentry.

• Processing location: United States (or as disclosed by Sentry).

5. Product Analytics (Current Status)

Product analytics (e.g. feature usage, session analytics, or tools such as PostHog) are not currently in use by AnalytIQ. No product analytics Subprocessor is active at this time.

If AnalytIQ adopts a product analytics provider in the future, it will be added to this list and disclosed in the Privacy Policy. Such use would be limited to product improvement and would not be used for advertising or cross-site behavioral profiling.

6. Subprocessor Management and Updates

• Subprocessors are evaluated on security practices, data protection commitments, and compliance with our contractual and legal obligations.

• This list is updated periodically as Subprocessors are added, removed, or changed.

• Objections to a new or changed Subprocessor may be raised in accordance with the Data Processing Agreement or your contract with AnalytIQ.

7. Data Protection Commitment

All Subprocessors are required to:

- Process data only on behalf of AnalytIQ and in accordance with our instructions.
- Maintain confidentiality and security safeguards appropriate to the data and the service.
- Comply with applicable privacy and education data protection laws (e.g. FERPA where applicable).
- Support AnalytIQ's FERPA compliance obligations when processing education records.

AnalytIQ remains responsible for ensuring Subprocessors meet these requirements.

8. Contact Information

Questions about Subprocessors or data handling may be directed to AnalytIQ through the contact information on the AnalytIQ website (e.g. www.myanalytiq.org) or at techsupport@myanalytiq.org or aelharouchi@myanalytiq.org.

AnalytIQ Acceptable Use Policy

AnalytIQ Acceptable Use Policy

Last Updated: February 20, 2026

This Acceptable Use Policy (“Policy”) describes the standards of conduct that apply to users of the AnalytIQ platform (the “Service”). This Policy is incorporated into the AnalytIQ Terms and Conditions. By accessing or using the Service, you agree to comply with this Policy.

1. Purpose of This Policy

AnalytIQ is designed to support learning, teaching, and educational use. This Policy ensures the platform is used in a safe, ethical, and lawful way, consistent with educational and institutional standards. Compliance with this Policy is a condition of use of the Service.

2. Permitted Use

You may use the Service only for legitimate educational and professional purposes, including:

- Participating in coursework and class activities.
- Creating and managing educational content (e.g. classes, chapters, exercises, resources).
- Practicing coding and completing assignments.
- Using AI-assisted learning support (e.g. chat, hints) as intended by the platform.
- Collaborating with instructors and classmates within the platform’s features.

Use must comply with applicable laws and institutional policies (including academic integrity and acceptable use).

3. Prohibited Activities

You may not use the Service to do any of the following.

3.1 Violate laws or regulations

- Engage in unlawful activities.
- Upload or distribute illegal content.
- Attempt to circumvent legal restrictions.

3.2 Compromise platform security

- Attempt unauthorized access to accounts, systems, or data (e.g. using another user’s credentials, bypassing authentication or authorization).
- Interfere with platform operations (e.g. circumventing rate limits, overloading systems, or disrupting availability).
- Introduce malware, harmful code, or exploits.

• Conduct vulnerability scanning or penetration testing without prior written authorization from AnalytIQ.

3.3 Misuse educational content

• Upload content that infringes intellectual property or other third-party rights.
• Share copyrighted materials without permission or outside authorized educational use.
• Distribute course materials outside the uses permitted by the platform and by your institution.

3.4 Harass or harm others

• Engage in harassment, bullying, or threatening behavior.
• Upload abusive or discriminatory content.
• Impersonate other users, institutions, or AnalytIQ.

3.5 Abuse AI features

You may not use AI tools to:

• Generate harmful, illegal, or inappropriate content.
• Circumvent academic integrity policies (e.g. using AI to complete work that must be your own under institutional rules).
• Attempt to exploit or reverse-engineer AI systems (e.g. probing for prompts, attempting to extract training data, or abusing rate or size limits).
• Use AI for non-educational automated or bulk activities (e.g. high-volume automated requests outside normal learning use).

3.6 Automated or unauthorized data collection

You may not:

• Scrape or extract platform data using automated tools (e.g. bots, scripts) without permission.
• Conduct bulk data harvesting or systematic collection of other users' data or content.
• Access data beyond what your role and permissions allow (e.g. accessing another instructor's classes or another student's progress without authorization).

4. Academic Integrity Expectations

AnalytIQ is an educational support platform and does not replace or override institutional academic policies.

You must:

- Follow your institution's academic integrity and honor code rules.
- Use AI assistance as a learning aid where permitted, and not as a substitute for original work where your institution requires original work.
- Avoid submitting AI-generated content as your own if your institution prohibits it.
- AnalytIQ does not monitor or enforce grading decisions or institutional honor codes.

Academic integrity enforcement remains the responsibility of educational institutions and instructors.

5. Data and Privacy Protection

You must not:

- Access or attempt to access other users' personal data without authorization.
- Share confidential educational records (e.g. progress, grades, submissions) outside the uses permitted by the platform and by your institution.
- Use the platform to store or process unrelated sensitive personal data (e.g. data that does not belong in an educational learning context).

6. Enforcement and Consequences

AnalytIQ reserves the right to investigate suspected violations of this Policy (e.g. using logs, audit trails, and reported information). If a violation occurs, AnalytIQ may, in its sole discretion:

- Issue warnings.
- Suspend or restrict account access (e.g. temporary or partial restriction).
- Remove offending content (e.g. classes, chapters, exercises, or resources that violate this Policy).
- Terminate accounts for serious or repeated violations.

AnalytIQ may use technical measures (e.g. rate limiting, IP blocklisting, account disable) to protect the platform and other users. AnalytIQ may notify educational institutions when violations involve academic integrity, institutional policies, or data that belongs to the institution.

7. Reporting Violations

You may report suspected violations of this Policy through the contact information provided on the AnalytIQ website (e.g. www.myanalytiq.org) or at techsupport@myanalytiq.org or aelharouchi@myanalytiq.org. Reports will be handled in accordance with AnalytIQ's internal procedures and applicable law.

8. Changes to This Policy

AnalytIQ may update this Policy from time to time. The “Last Updated” date at the top will be revised when changes are made. Material changes may be communicated via the Service or by other reasonable means. Continued use of the Service after the effective date of an update constitutes acceptance of the revised Policy. If you do not agree, you must stop using the Service.

AnalytIQ Security and Technical

Safeguards

AnalytIQ Security Overview and Technical Safeguards

Last Updated: February 20, 2026

1. Introduction

AnalytIQ is committed to protecting the confidentiality, integrity, and availability of user data, including educational records processed on behalf of institutions. This document describes the administrative, technical, and physical safeguards used to secure the AnalytIQ platform. These safeguards are designed to align with industry standards for cloud-based educational technology services.

2. Security Principles

AnalytIQ's security approach is based on:

• Least-privilege access — Users and systems can access only the data and functions required for their role or task.

• Secure-by-default design — Security controls (e.g. encryption in transit, role-based access) are built into the application and hosting stack.

• Strong encryption — Data is protected in transit (TLS) and at rest (infrastructure-level encryption; additional application-level encryption for sensitive configuration where appropriate).

• Ongoing monitoring and logging — Application and security-relevant events are logged and, where deployed, monitored for errors and anomalies.

• Separation of data — Educational and user data are stored in dedicated database and storage systems with access controlled by role and tenant; operational data (e.g. logs, caches) are separated from persistent personal and educational records where feasible.

3. Infrastructure Security

3.1 Cloud Hosting Environment

AnalytIQ runs on managed cloud infrastructure in the United States, including:

• Application hosting (e.g. Render) for the web application and optional background workers.

• Managed PostgreSQL database and object storage (e.g. Supabase) for all persistent application data and stored files.

• Redis (e.g. via Render or equivalent) for rate limiting, caching, and (when SSO is enabled) OAuth2 state; Redis is not used for persistent storage of personal or educational records. Supabase may use underlying cloud infrastructure (e.g. AWS) for database and storage. AnalytIQ does not operate its own data centers; physical and environmental security are provided by these infrastructure providers.

3.2 Network Security

The platform enforces:

• HTTPS (TLS) for all user-to-platform and platform-to-third-party traffic.

• Secure TLS configurations for encrypted transmission.

• Provider-level network and firewall controls for hosting and database services (e.g. restricted access, network isolation as provided by Render, Supabase, and underlying infrastructure).

3.3 Data Isolation

Logical separation of data is enforced by the application:

• Tenant and role boundaries — Users can access only data that belongs to their tenant (where applicable) and that their role is allowed to see (e.g. students see their own data and enrolled classes; instructors see their classes and related students).

• Authorization checks — Every API request is authenticated and authorized; there is no direct client access to the database. Data access is restricted to authorized contexts based on institutional and role permissions.

4. Application Security

4.1 Secure Authentication

AnalytIQ uses:

• Secure password hashing (e.g. bcrypt with a configurable work factor); plain-text passwords are not stored.

• JSON Web Token (JWT) session authentication; tokens are signed, include role and (where applicable) tenant scope, and have a configurable expiration.

• Session expiration and validation on each request; expired or invalid tokens are rejected and the client is directed to re-authenticate.

• Optional per-tenant SSO (OAuth2/OIDC) and LTI 1.3 for LMS-based access, with state and signature validation to protect against misuse.

4.2 Access Controls

Access to data is restricted through:

• Role-based permissions (e.g. student, professor, tenant_admin, superadmin) enforced by the backend on every API call.

• Authorization checks at the API level (e.g. class ownership, enrollment, tenant scope).

• Enforcement of class and enrollment boundaries so users cannot access data outside their authorized roles or classes.

• Users cannot access data outside their authorized roles or institutional scope.

4.3 Input Validation and Protection

AnalytIQ reduces common web risks through:

• Parameterized database queries for all SQL to prevent injection.

• Input validation and length limits for user-supplied data (e.g. names, emails, passwords, class and chapter metadata, exercise content, AI messages), with limits defined in configuration.

• Optional CSRF protection when enabled (e.g. for form-based flows); API authentication is token-based (Bearer JWT).

• Security headers (e.g. X-Content-Type-Options, X-Frame-Options, Referrer-Policy; HSTS in production) and rate limiting on authentication and sensitive endpoints.

5. Encryption and Data Protection

5.1 Encryption in Transit

• All data transmitted between users and the platform, and between the platform and third-party services, is encrypted using HTTPS and TLS.

5.2 Encryption at Rest

• Data stored in the database and object storage is encrypted at rest using the encryption provided by the infrastructure providers (e.g. Supabase).

• Sensitive configuration (e.g. SSO client secrets) is additionally encrypted by the application (e.g. symmetric encryption) before being stored in the database.

5.3 Credential Protection

• Passwords are stored only in hashed form (e.g. bcrypt) and are never transmitted or

stored in plain text.

• Session tokens (JWTs) are signed and validated server-side; they are not stored on the server after issuance.

• Third-party credentials (e.g. SSO client secrets) are encrypted before storage and are not logged or exposed to clients.

6. AI Data Security

AnalytIQ applies safeguards to AI-related processing:

6.1 Processing and Retention

• AI prompts and responses are used only to deliver the feature (e.g. chat, hints).

Conversation data may be stored in the platform's database to support multi-turn chat and continuity; such storage is for service delivery only and is not used for model training. Only the data necessary to generate a response (e.g. current prompt and permitted context) is sent to the AI provider per request.

6.2 No AI Training Use

• Student data and educational records are never used to train artificial intelligence models (by AnalytIQ or the AI provider for training purposes).

6.3 Limited Data Exposure

• Only the minimal data required to generate responses is transmitted to the AI service provider (e.g. Google Gemini API). Rate and size limits apply to reduce abuse and limit exposure.

7. Monitoring and Logging

AnalytIQ maintains logging to support security and operations:

• Structured application logs (e.g. JSON with request identifiers) for requests, errors, and security-relevant events.

• Audit logging in the database for designated user and data mutations (e.g. account and content changes) to support accountability and incident response.

• Optional error and performance monitoring (e.g. Sentry) with PII filtered or removed before data is sent to the monitoring provider.

• Request tracing (e.g. request IDs in logs and responses) to support incident investigation.

Logs are retained only as long as necessary for operational and security purposes, in line with the Data Retention and Deletion Policy and provider practices.

8. Incident Response

AnalytIQ maintains procedures to respond to security incidents, as set out in the Incident

Response Policy. These include:

- Detection and investigation using logs, audit trails, and reported information.
- Containment and remediation (e.g. IP blocklisting, account disable, rate limiting, patching, configuration changes).
- Notification to affected institutions when required by law or contract (e.g. after a confirmed incident involving personal or educational data).
- Documentation and review of incidents for lessons learned and improvement of controls.

9. Access Management

9.1 Internal Access Controls

Access to production systems (e.g. hosting, database, and operational tools) is limited to authorized personnel with a legitimate operational need and is governed by AnalytIQ's internal policies and provider access controls.

9.2 Administrative Privileges

Administrative access within the platform (e.g. tenant_admin, superadmin roles) is restricted, enforced by role, and logged in the audit trail. Administrative actions are subject to the same authentication and authorization controls as other users.

9.3 Super Administrator Controls

Super administrator functions are restricted to users with the superadmin role, require strong authentication (same as other users; no separate MFA at this time), and administrative actions are recorded in the audit log for accountability.

10. Subprocessor Security

AnalytIQ uses only trusted subprocessors (e.g. Supabase, Render, Google Gemini, email provider, error monitoring) that are expected to implement appropriate security safeguards.

Subprocessors are contractually required to:

- Protect user data in line with their role and our instructions.
- Maintain confidentiality and comply with applicable data protection requirements.
- Use data solely to provide the agreed services.
- A current list is maintained in the AnalytIQ Subprocessor List.

11. Data Backup and Recovery

Backups of the database and, where applicable, other backed-up data are managed by the infrastructure and database provider (e.g. Supabase) or the hosting provider.

These providers typically offer:

- Automated backups and retention schedules.
- Data redundancy and availability controls.
- Disaster recovery and business continuity capabilities.

AnalytIQ does not operate its own backup systems; recovery is dependent on provider capabilities and retention cycles. Upon deletion requests or contract termination, active data is deleted or returned, and backup data is purged through the provider's normal retention and overwrite cycles.

12. Compliance Alignment

AnalytIQ's security practices are designed to support compliance with:

- FERPA (education privacy).
- Applicable U.S. privacy laws (e.g. state breach notification, CCPA/CPRA where applicable).
- Industry standards for cloud software security (e.g. encryption, access control, logging, incident response).

13. Security Responsibilities of Institutions

Educational institutions are responsible for:

- Managing their internal access policies and user accounts (e.g. LMS, SSO).
- Protecting institutional credentials and systems that integrate with AnalytIQ.
- Maintaining security within their own systems and communicating with students and staff as required (e.g. after a breach notification). AnalytIQ supports institutional security through the platform safeguards described in this document and through contractual commitments (e.g. DPA, incident notification).

14. Policy Updates

This document may be updated periodically to reflect changes in security practices, infrastructure, or legal requirements. The "Last Updated" date at the top will be revised when changes are made.

15. Contact for Security Matters

Security inquiries may be directed to AnalytIQ through the contact information provided on the AnalytIQ website (e.g. www.myanalytiq.org) or at techsupport@myanalytiq.org or aelharouchi@myanalytiq.org.

AnalytIQ Responsible AI and AI

Transparency Policy

AnalytIQ Responsible AI and AI

Transparency Policy

Last Updated: February 20, 2026

1. Purpose

AnalytIQ uses artificial intelligence to support learning, teaching, and educational content creation. This policy describes how AnalytIQ designs, deploys, and governs AI features in a responsible, ethical, and privacy-protective way. It is intended to align with applicable law, institutional expectations, and evolving AI governance standards.

2. Role of AI in the Platform

AI in AnalytIQ is used only to support educational use, including:

• Contextual coding assistance (e.g. chat and hints in response to user questions and code).

• Generating practice exercises from educational materials (e.g. chapter text) for instructors to review and publish.

• Explaining concepts and helping debug code in chat and hint interactions.

• Supporting learning through on-demand assistance and content generation where the platform offers it. AI features are designed to assist learning and teaching, not to replace human instruction or academic decision-making (e.g. grading policy, course design, or institutional standards).

3. Human Oversight

AnalytIQ ensures AI is used with human oversight:

• Instructors review and approve (or edit) AI-generated exercises before they are used by students; AI does not publish content directly to students without instructor control.

• Students remain responsible for their own work and decisions; AI is a support tool, not the author of their submissions.

• Institutions keep control over grading, academic integrity, and academic policies; AnalytIQ does not set or enforce those policies. AI does not make independent academic judgments (e.g. final grades or eligibility decisions). Grading of student code is performed by server-side execution against AI-generated unit test cases. AI generates the test cases during exercise creation (under instructor review and approval); AI does not perform real-time grading decisions during student submission. Scores are stored on the backend.

4. Data Use and Privacy Protections

4.1 No Training on Student Data

AnalytIQ does not use the following to train artificial intelligence models:

- Student submissions or coursework.
- Educational records (e.g. progress, enrollments, scores).
- User prompts or AI conversation content.

This applies to AnalytIQ and to the AI service provider (e.g. Google Gemini) for training; data sent to the provider is used only to generate responses for the user.

4.2 Processing and Retention

- AI inputs (e.g. prompts, context) are processed to generate real-time responses.
- Conversation data (e.g. messages) may be stored in the platform's database to support

multi-turn chat and continuity; such storage is for service delivery only and is not used for model training or long-term retention beyond what is needed for the feature.

- Retention of AI-related data follows the same retention and deletion rules as other user

data (e.g. removed or anonymized when the account or related data is deleted, where applicable).

4.3 Minimal Data Sharing

- Only the minimum data necessary to generate a response (e.g. current message and

permitted context, within size and length limits) is sent to the AI service provider per request. Rate limits and message/size caps apply to reduce abuse and limit data exposure.

5. FERPA Alignment

When the platform is used by educational institutions, AI features operate within FERPA protections. AnalytIQ acts as a school official with a legitimate educational interest; AI processing is done only to provide the educational services requested by the institution and in accordance with the Data Processing Agreement and FERPA Compliance Statement.

Student data processed by AI is not used for advertising, sale, or model training.

6. Accuracy and Limitations of AI

AI outputs may:

- Contain inaccuracies or errors.
- Be incomplete or outdated.
- Require human verification before being relied upon.

AI assistance is intended as a learning and teaching aid, not as a sole source of truth or professional, legal, or academic advice. Users should evaluate and verify AI-generated content, especially when it affects grades, submissions, or institutional decisions.

7. Academic Integrity and Ethical Use

AnalytIQ encourages ethical use of AI in education.

Users must:

- Follow their institution's academic integrity and honor code policies.
- Use AI assistance to support learning rather than to replace original work where the institution requires original work.
- Avoid submitting AI-generated content as their own when institutional rules prohibit it.

Institutions remain responsible for defining and enforcing academic policies; AnalytIQ does not monitor or enforce institutional academic integrity rules.

8. Safety and Risk Mitigation

AnalytIQ uses safeguards to reduce AI-related risks, including:

- Rate limiting (e.g. per-user per minute and per hour) on AI chat to limit abuse and resource use.
- Payload and message limits (e.g. message size and conversation length caps) so requests stay within intended use.
- Operational monitoring (e.g. error rates, circuit breaker on repeated AI provider failures) to support reliability and detection of issues.
- Contractual and technical measures with the AI provider (e.g. Google Gemini) so that data is used only for generating responses and not for training.
- AnalytIQ relies on the AI provider's safety and content policies where applicable; the platform does not independently filter or moderate all AI outputs beyond the controls above.

9. Transparency to Users

AnalytIQ clearly indicates when AI is being used (e.g. in chat and hint features).

Users are informed (through the product and policies) that:

- AI responses are machine-generated and may be incomplete or wrong.
- AI outputs may require verification and should not be relied upon as the sole source of truth.
- AI is for educational assistance only and does not replace instructor judgment or institutional policy.

10. Continuous Improvement

AnalytIQ periodically evaluates AI systems to improve:

- Reliability (e.g. availability, error handling).
- Safety (e.g. rate limits, provider safeguards).
- Educational value (e.g. usefulness of chat and exercise generation).
- Fairness and bias where relevant (e.g. in how AI features are used and documented).

Updates to AI features, safeguards, and this policy are made as technology, regulation, and governance standards evolve.

11. Third-Party AI Providers

AnalytIQ uses trusted AI service providers (e.g. Google Gemini API) that are expected to maintain strong privacy and security practices.

These providers:

- Process data only to deliver the requested functionality (e.g. generate a response).
- Do not use AnalytIQ user data for training their models (as contractually and technically

arranged).

- Are subject to contractual data protection and confidentiality requirements.

A current list of subprocessors, including the AI provider, is maintained in the AnalytIQ Subprocessor List.

12. Policy Updates

This policy may be updated from time to time to reflect changes in technology, regulation, or AI governance standards. The “Last Updated” date at the top will be revised when changes are made. Continued use of AI features after an update constitutes acceptance of the revised policy.

13. Contact for AI Governance Matters

Questions about AI use, this policy, or AI governance may be directed to AnalytIQ through the contact information on the AnalytIQ website (e.g. www.myanalytiq.org) or at techsupport@myanalytiq.org or aelharouchi@myanalytiq.org.

A nalytIQ IncidentandDataBreach Policy

AnalytIQ Incident Response and Data

Breach Notification Policy

Last Updated: February 20, 2026

1. Purpose

This policy defines how AnalytIQ detects, investigates, responds to, and communicates security incidents, including potential data breaches involving personal data or education records.

Goals:

- Protect user data and system integrity.
- Minimize the impact of security incidents.
- Notify affected institutions in a timely way when required.
- Support compliance with FERPA and applicable data protection and breach-notification laws.

2. Scope

This policy applies to all security incidents involving:

- Personal data of users (e.g. account information, progress, submissions).
- Student education records processed on behalf of institutions.Platform infrastructure and systems (e.g. application, database, storage, third-party integrations).
- Subprocessors that process data on AnalytIQ's behalf.

3. Definition of a Security Incident

A security incident is any event that may compromise:

- Confidentiality of data (e.g. unauthorized access or disclosure).
- Integrity of systems or records (e.g. unauthorized modification or destruction).
- Availability of the Service (e.g. significant disruption or denial of service).

Examples include:

- Unauthorized access to user accounts or administrative interfaces.
- Data exposure due to misconfiguration, vulnerability, or breach.
- Malware, ransomware, or other malicious activity.
- Service disruptions caused by cyber threats or abuse.

• Accidental data disclosure (e.g. to the wrong recipient or via a logged error).

• Abuse of the platform (e.g. scraping, credential stuffing, or AI endpoint abuse) that poses a material risk to users or data.

4. Incident Response Principles

AnalytIQ follows these principles when responding to incidents:

• Rapid detection and containment to limit harm and exposure.

• Accurate assessment of risk, impact, and affected data (e.g. types of data, number of users or records).

• Transparent communication with affected institutions when notification is required by law or contract.

• Continuous improvement after resolution (e.g. post-incident review, updates to controls and procedures).

5. Incident Detection

Security incidents may be identified through:

• Application and security logging (e.g. structured logs with request identifiers, errors, and security-relevant events).

• Audit trail (e.g. user and data mutations recorded in the database).

• Error and performance monitoring (e.g. Sentry or equivalent, with PII minimized) where deployed.

• Infrastructure or provider alerts (e.g. from Render, Supabase, or other subprocessors).

• Reports from users or institutions (e.g. via the platform's contact channel or support).

• Notifications from subprocessors regarding incidents affecting data they process on AnalytIQ's behalf. All suspected incidents are reviewed promptly to determine whether they meet the definition of a security incident and require formal response and, where applicable, notification.

6. Incident Response Process

AnalytIQ follows a structured response process (described in full in the Incident Response

Policy). Summary of stages:

6.1 Identification

The responsible team evaluates alerts, reports, or logs to confirm whether an incident has

occurred and to determine:

• Scope (e.g. which systems, environments, or data stores).

• Severity (e.g. impact on confidentiality, integrity, or availability).
• Types of data potentially affected (e.g. account data, education records).

6.2 Containment

Immediate actions are taken to limit impact, which may include:

• Restricting or revoking access (e.g. disabling compromised or abusive user accounts, forcing password reset where appropriate).

• Blocking abusive or malicious traffic (e.g. using platform-level controls such as IP blocklisting where supported).

• Rate limiting or throttling to reduce ongoing abuse (e.g. of authentication or AI endpoints).

• Coordination with infrastructure or subprocessors (e.g. to isolate affected systems or accounts at the provider level, apply emergency patches, or revoke credentials).

• Containment continues until the threat or exposure is under control.

6.3 Investigation

A detailed investigation is conducted to determine:

• Root cause of the incident.
• Systems and data affected.
• Duration of exposure or compromise (if reasonably determinable).
• Risk level to users and institutions (e.g. likelihood of misuse of exposed data).

6.4 Remediation

Corrective actions are implemented to:

• Fix identified vulnerabilities or misconfigurations (e.g. patches, configuration changes, credential rotation).

• Strengthen security controls (e.g. access controls, logging, validation).

• Reduce the risk of recurrence.

6.5 Recovery

Affected systems are restored to normal operation after confirming that security and stability are sufficient. Additional monitoring may be put in place for a period. Post-incident review (Section 10) is conducted to capture lessons learned and improve future response.

7. Data Breach Notification

7.1 Notification to Institutions

If a confirmed incident involves unauthorized access to or disclosure of personal data or education records, AnalytIQ will notify affected institutions without undue delay after the incident is confirmed and key facts are known. Notification will include, to the extent known and legally permissible:

- Description of the incident (e.g. nature of the event, how it was discovered).
- Types of data affected (e.g. account information, education records, progress data).
- Estimated number of impacted users or records, if reasonably determinable.
- Steps taken to contain the incident and mitigate harm.
- Recommended actions for the institution (e.g. user communication, credential resets, monitoring).

AnalytIQ will cooperate with institutions in meeting any legal or regulatory reporting obligations (e.g. breach notification to authorities or data subjects) that apply to the institution or to AnalytIQ as processor.

7.2 FERPA Compliance

When education records are involved, AnalytIQ works directly with the educational institution.

The institution remains responsible for:

- Notifying students and, where required, regulators in accordance with FERPA and applicable state or federal breach laws.
- Deciding the content and timing of any notice to students or parents.

AnalytIQ provides the institution with the information needed to fulfill these obligations (consistent with Section 7.1) and assists as required by contract or law.

7.3 Notification to Individuals

AnalytIQ typically does not notify students or other end users directly unless:

- Required by law (e.g. state breach law imposing a direct notification duty on the processor), or
- Instructed by the institution (e.g. the institution asks AnalytIQ to send notices on its behalf).

When the institution is the data controller, notification to data subjects is coordinated with the institution so that messaging is consistent and compliant with the institution's obligations.

8. Subprocessor Incident Coordination

AnalytIQ requires subprocessors (by contract) to:

- Maintain incident response procedures appropriate to the services they provide.
- Notify AnalytIQ promptly of security incidents that may affect data processed for

AnalytIQ or the availability or security of the platform.

• Cooperate in investigations and remediation (e.g. providing information, applying patches, or implementing containment at the provider level).

• When an incident originates at or involves a subprocessor, AnalytIQ will assess impact on user and institutional data, contain exposure (including through the subprocessor), and notify affected institutions in accordance with Section 7.

9. Incident Documentation

All security incidents are documented, including:

• Timeline of events (detection, assessment, containment, remediation, recovery).

• Impact assessment (e.g. systems and data affected, severity, number of users or records if known).

• Remediation steps taken.

• Lessons learned and any follow-up actions (e.g. control improvements, policy updates).

• Documentation is retained for internal security review, compliance, and continuous improvement, in line with legal and contractual retention requirements.

10. Post-Incident Review

After resolution, AnalytIQ conducts a post-incident review to:

• Identify security and process improvements (e.g. detection, containment, communication).

• Update policies, procedures, or controls where needed.

• Strengthen monitoring, logging, or access controls to reduce the risk of recurrence.

Findings are used to improve the Incident Response Policy, this policy, and related runbooks or safeguards.

11. Preventive Security Measures

AnalytIQ reduces incident risk through preventive measures, including:

• Encrypted communications (HTTPS/TLS) for all user and third-party traffic.

• Secure cloud infrastructure (e.g. managed database and storage, application hosting) with provider-level security controls.

• Access controls and authentication (e.g. role-based access, strong password hashing, token-based sessions, optional SSO and LTI with validation).

• Application security (e.g. parameterized queries, input validation, rate limiting, security headers).

• Logging and audit trails (e.g. structured logs, audit log for mutations) to support detection and investigation.

• Ongoing monitoring (e.g. application and error monitoring where deployed) and regular dependency and security updates as part of the development and release process.

12. Institutional Responsibilities

Educational institutions remain responsible for:

• Maintaining secure access credentials and managing their own user accounts and LMS or SSO systems that integrate with AnalytIQ.

• Notifying students, parents, or regulators when required by law (e.g. FERPA, state breach laws) following a breach that affects their data.

• Managing security and incident response within their own systems and communicating with their communities as appropriate. AnalytIQ supports institutions by providing timely incident and breach information and assisting with notification and remediation where required by contract or law.

13. Policy Updates

This policy may be updated periodically to reflect changes in security practices, infrastructure, regulatory requirements, or contractual obligations. The “Last Updated” date at the top will be revised when changes are made.

14. Contact for Security Incidents

Security concerns or suspected incidents may be reported to AnalytIQ through the contact information on the AnalytIQ website (e.g. www.myanalytiq.org) or at techsupport@myanalytiq.org or aelharouchi@myanalytiq.org. Reports will be handled in accordance with this policy and the Incident Response Policy.

AnalytIQ Business Continuity and Disaster Recovery

AnalytIQ Business Continuity and Disaster Recovery Policy

Last Updated: February 20, 2026

1. Purpose

This policy describes how AnalytIQ maintains service availability and restores operations after system failures, cyber incidents, or infrastructure disruptions.

Objectives:

- Maintain continuity of the AnalytIQ platform for users and institutions.
- Protect user and education data from loss (through provider-managed backups and durable storage).
- Minimize downtime and service disruption (through managed infrastructure and recovery procedures).
- Support reliable access for educational institutions consistent with provider capabilities and this policy.

2. Scope

This policy applies to systems that support the AnalytIQ platform, including:

- Application servers (e.g. web application and optional background workers hosted on Render or equivalent).
- Databases (managed PostgreSQL, e.g. via Supabase).
- Storage systems (object storage for course materials and uploaded files, e.g. via Supabase).
- Caching and background processing (e.g. Redis for rate limiting, caching, SSO state, and optional task queues).
- Third-party infrastructure providers (e.g. Render, Supabase, and their subprocessors) that host or process platform data.

3. Hosting and Infrastructure Overview

AnalytIQ runs on managed cloud infrastructure in the United States, including:

- Application hosting (e.g. Render) for the web application and optional background workers (e.g. Celery when enabled).
- Managed PostgreSQL database (e.g. Supabase) for all persistent application data.
- Cloud object storage (e.g. Supabase Storage) for course materials, uploaded files, and other stored assets.

• Redis (e.g. via Render or equivalent) for caching, rate limiting, and (when SSO is enabled) OAuth2 state; and, when async tasks are enabled, for background task queuing. AnalytIQ does not operate its own data centers or physical infrastructure. Physical and network security, redundancy, and availability are provided by these infrastructure providers, which maintain enterprise-grade controls and reliability commitments.

4. Business Continuity Strategy

AnalytIQ supports service continuity through:

• Managed cloud infrastructure with provider-level redundancy, monitoring, and failover where offered (e.g. database and storage durability, application hosting availability).

• Automated monitoring and alerting (e.g. application and error monitoring, provider status and alerts) to detect failures and support response.

• Provider reliability and availability commitments (as set out in provider terms and SLAs).

• Deployment and rollback (e.g. version-controlled code, CI/CD, ability to deploy or roll back application versions) to restore or stabilize service after code-related issues. The platform is designed to rely on provider redundancy and recovery capabilities; localized failures (e.g. a single component or availability zone) may be mitigated by the provider without complete platform outage, depending on provider design and incident scope.

5. Backup Procedures

5.1 Database Backups

The primary database (PostgreSQL) is backed up by the database provider (e.g. Supabase), not by AnalytIQ directly.

Provider backup practices typically include:

• Automated backups on a schedule (e.g. daily or more frequent, as per provider).

• Point-in-time recovery (where offered) to restore to a chosen point within the retention window.

• Encrypted storage of backup data and retention according to the provider’s policy.

Exact backup frequency, retention, and RPO are determined by the provider; AnalytIQ does not operate its own database backup system.

5.2 File Storage Backups

Course materials and uploaded files are stored in cloud object storage (e.g. Supabase Storage).

The provider typically provides:

- Redundancy and durability (e.g. replication, durability guarantees).
- Availability and recovery capabilities in line with the provider's design and SLA.

AnalytIQ does not run separate backup systems for file storage; continuity and recovery depend on provider redundancy and backup/recovery features.

5.3 Application Code and Configuration

- Platform code is stored in version-controlled repositories (e.g. Git/GitHub) with:
- Full revision history and the ability to restore previous stable versions.
- Redundant storage and access controls offered by the repository provider.

Configuration and secrets (e.g. environment variables, keys) are managed via deployment and hosting (e.g. Render) and are not backed up in the same way as user data; recovery of application and configuration is achieved through redeployment and provider recovery where applicable.

6. Recovery Objectives

AnalytIQ aims for recovery standards consistent with provider capabilities and operational constraints. Actual recovery times and data loss depend on provider backup schedules, recovery procedures, and incident scope. Recovery Time Objective (RTO)

- Target: Restore service within 24 hours or less for major system disruptions, where

restoration is within AnalytIQ's control (e.g. application redeployment, configuration correction). For provider-level outages (e.g. database or region failure), recovery time depends on the provider's restoration and may exceed 24 hours. Recovery Point Objective (RPO)

- Target: Minimize data loss; in practice, RPO is determined by provider backup frequency

(e.g. daily or more frequent database backups). AnalytIQ does not guarantee a specific RPO; the objective is to align operations and provider selection with less than 24 hours of potential data loss where feasible.

7. Disaster Recovery Process

In the event of a major system failure or disaster, AnalytIQ follows a structured recovery process (aligned with the Incident Response Policy and Incident Response and Data Breach Notification Policy where the event is a security incident).

7.1 Incident Assessment

- The responsible team evaluates the nature and scope of the disruption (e.g. application

failure, database unavailability, storage or provider outage) and identifies affected systems and dependencies.

7.2 Containment and Isolation

• Compromised or malfunctioning components are contained or isolated to prevent further impact. Where the failure is within provider infrastructure, AnalytIQ coordinates with the provider (e.g. support, status updates) and may isolate or disable platform features that depend on the affected service until recovery is complete.

7.3 Restoration

• Systems and data are restored using provider-managed backups and recovery procedures (e.g. database restore from backup, storage recovery, application redeployment from version control). AnalytIQ does not perform its own low-level backup restore; restoration is carried out through the infrastructure providers and redeployment of the application where needed.

7.4 Verification and Testing

• Restored systems are verified and tested (e.g. connectivity, critical flows, data integrity checks where feasible) before full service resumption or before declaring the incident resolved.

7.5 Communication

• Affected institutions may be notified when service disruptions materially impact platform availability (e.g. extended outage, data impact). Communication is consistent with the Incident Response and Data Breach Notification Policy when the event involves a security incident or data breach.

8. High Availability Measures

AnalytIQ reduces downtime risk through:

• Managed cloud hosting (e.g. Render, Supabase) with provider-level redundancy and availability.

• Redundant or durable storage (e.g. database and object storage) as provided by the infrastructure provider.

• Automated monitoring and alerting (e.g. application health checks, error monitoring, provider status) to detect issues and support rapid response.

• Secure configuration management (e.g. environment-based configuration, secrets management via the hosting platform) and regular dependency and security updates. These measures are intended to support reliable platform performance; actual availability depends on provider uptime and the scope of any incident.

9. Subprocessor Continuity

Third-party providers (e.g. Render, Supabase) that support the platform maintain their own disaster recovery and business continuity procedures and commitments (e.g. SLAs, status pages, incident communication).

AnalytIQ relies on these provider commitments to:

- Maintain infrastructure uptime and availability.
- Protect stored data (e.g. backups, durability, encryption).
- Provide recovery and restoration capabilities in the event of failure.

A current list of subprocessors is maintained in the AnalytIQ Subprocessor List. In the event of a provider incident, AnalytIQ will assess impact, coordinate with the provider, and communicate with institutions as appropriate (see Section 7.5).

10. Testing and Review

Business continuity and disaster recovery practices (e.g. recovery procedures, provider dependencies, communication plans) are reviewed periodically to ensure they remain effective and aligned with operational needs, provider capabilities, and any changes to the platform or infrastructure.

11. Limitations

Although AnalytIQ implements reasonable safeguards (as described in this policy and related security documentation), no system can guarantee uninterrupted service or complete protection against all disasters, cyber incidents, or provider failures.

Recovery times and data loss depend on provider backup schedules, recovery procedures, and the nature and scope of each incident. RTO and RPO are objectives, not contractual guarantees, unless otherwise agreed in a separate service agreement.

12. Policy Updates

This policy may be updated periodically to reflect infrastructure changes, provider changes, and evolving best practices or regulatory expectations. The “Last Updated” date at the top will be revised when changes are made.

13. Contact

Questions about business continuity or disaster recovery may be directed to AnalytIQ through the contact information on the AnalytIQ website (e.g. www.myanalytiq.org) or at techsupport@myanalytiq.org or aelharouchi@myanalytiq.org.